

Tutorial

Workflow for the Calculation, Analysis and Mapping of Marine Heatwaves (MHWs) and Marine Cold Spels (MCSs) in Xraster

November 2026

Marine Heatwaves (MHWs)

A **marine heatwave (MHW)** is defined as a period during which the local sea temperature exceeds a locally determined threshold, typically the **90th percentile of the local long-term climatology**, for at least **five consecutive days** (Hobday et al., 2016).

A marine **cold spell (MCS)** is defined as a period during which the local sea temperature remains below a locally determined threshold, typically the 10**th percentile of the local long-term climatology**, for at least **five consecutive days.**

The methods used to identify and analyse MHWs and MCSs are analogous and can also be applied to variables other than temperature.

The long-term climatology is the statistics of **temperature at a given location for each calendar day (day of year)**, calculated over a sufficiently long reference period. In this workflow, the climatological statistics (mean, 90th percentile and 10th percentile) for each calendar day is calculated using an **11-day moving window (±5 days)** centered on that day, with the corresponding days from all years of the reference period included in the calculation. A period of **at least 30 years** is generally preferred to obtain a robust long-term climatology.

MHW intensity metrics

Oliver et al. (2018) introduced several measures for characterizing MHW intensity and duration:

- **Duration** – number of days of the MHW (days)
- **Frequancy** - number of individual Marine Heatwave events occurring at a given location or within a defined area of interest during a specified period
- **Maximum intensity** – maximum temperature anomaly above the climatology (°C)
- **Mean intensity** – mean temperature anomaly above the climatology (°C)
- **Cumulative intensity** – sum of the daily temperature anomalies above the climatology (°C·days)

The standard MHW definition is generally applied **independently at each spatial grid cell**. By combining the affected grid cells, additional measures describing the spatial extent of an MHW can be calculated:

- **Maximum spatial extent** – the largest area affected during an event (km^2)
- **Mean spatial extent** – the mean area affected during an event (km^2)
- **Cumulative area** – the sum of the daily affected areas over the duration of the event (km^2·days)
- **Relative spatial coverage** – percentage of the study area affected by the MHW (%)

MHW categories

An MHW can also be assigned a **category** based on its intensity. The MHW category framework proposed by Hobday et al. (2018) classifies events according to how far the temperature exceeds the **90th percentile threshold**, relative to the local climatology.

The categories are:

- **Moderate** – Category I
- **Strong** – Category II
- **Severe** – Category III
- **Extreme** – Category IV

The category assigned to an MHW event is usually determined by the **highest category reached during any day of the event**. For example, if an event is Moderate for several days but reaches Strong intensity on one day, the event is classified as a **Strong MHW**.

This categorization provides a standardized framework for comparing the severity of MHWs between regions with different background climates. It was introduced by Hobday et al. (2018) and is now widely used in marine heatwave research and operational monitoring.

MHWs as spatio-temporal objects

MHWs can also be treated as **three-dimensional spatio-temporal objects**, with two spatial dimensions and one temporal dimension (longitude × latitude × time). In this approach, an MHW is defined as a **coherent spatio-temporal object formed by the connectivity of anomalously warm grid cells in space and time**.

The object-based approach builds on the local MHW threshold definition. The principal characteristics of an MHW treated as a spatio-temporal object are:

- **duration** – the temporal extent of the object;
- **spatial extent** – the area occupied by the object, which may change over time;
- **intensity** – the magnitude of the temperature anomaly;
- **category** – the MHW intensity category assigned to the object.

Existing tools for MHW analysis

Currently, tools for calculating marine heatwaves are mainly specialized programming packages based on the standard Hobday et al. definition: temperatures exceeding the local 90th-percentile threshold for at least five days.

Commonly used tools include the R package *heatwaveR* , the Python package *marineHeatWaves*, and spatial tracking tools such as *Ocetrac*.

These tools and frameworks can be broadly divided according to how they treat the spatial dimensions of the data.

Pixel-based tools, such as *heatwaveR* and *marineHeatWaves*, calculate MHW statistics independently for each spatial grid cell. For a gridded dataset, the calculation is repeated for every grid cell to identify periods when the local temperature threshold is exceeded.

Object-based tools, such as *Ocetrac*, *TempestExtremes*, and *m_mhw*, consider the spatial field as a whole. They identify connected groups of anomalously warm grid cells and track these groups through space and time. In this way, an MHW can be represented as a coherent three-dimensional structure (longitude × latitude × time) that can change its shape, area, and position, and can potentially split or merge.

Technical and practical limitations

The use of existing MHW tools can involve several technical and practical limitations:

1. **Memory and CPU requirements**
   Pixel-based calculations, such as those implemented in *heatwaveR* or *marineHeatWaves*, require the calculation of local thresholds and MHW events independently for potentially millions of grid cells over decades of daily data. Processing large global datasets can therefore require substantial computational resources and memory.

2. **The 3D big-data problem**
   Object-based approaches such as *Ocetrac* require processing large three-dimensional data cubes (longitude × latitude × time) to identify and track connected spatial structures. Large datasets may therefore require substantial memory and, in some cases, high-performance computing (HPC) or cloud-computing resources.

3. **Programming requirements**
   There is no widely adopted, user-friendly point-and-click **GUI** specifically designed for complete MHW analysis. Many existing workflows require programming skills in R or Python and familiarity with data-processing libraries such as `xarray`, `dask`, and `netCDF4`.

Xraster provides a GUI-based workflow for MHW analysis and mapping directly within QGIS, allowing gridded oceanographic data in NetCDF format to be processed and visualized without requiring the user to develop a complete programming workflow.

The MHW analysis workflow in Xraster consists of **four main stages**:

1. Data preparation

The first stage involves obtaining a **continuous daily sea-temperature dataset**, *T(t, x, y)*, on a regular spatial grid covering the entire study period, including a defined reference period of approximately **30 years**.

The data are prepared in **NetCDF format**, which is a standard format for multidimensional oceanographic data, such as datasets available from the Copernicus Marine Data Store.

2. Calculation of MHW variables and masks

The second stage involves calculating the following variables:

- **temperature anomaly**;
- **MHW category**;
- **MHW object ID**;
- **MHW object mask**.

The MHW object mask is defined according to the selected criteria for **object duration and spatial extent (area)**.

An **AOI (Area of Interest) mask** is also created to define the spatial area included in the analysis.

3. Creation of the MHW dataset and object descriptions

In the third stage, a new dataset is created containing the original **sea temperature** and the calculated MHW variables.

In addition, a **text file** is generated containing information describing individual MHW objects, including their temporal and spatial characteristics.

4. Visualization and analysis

The final stage consists of the **visualization and analysis of MHWs in Xraster**, using the created dataset and the accompanying text file.

The workflow starts by loading a **daily temperature NetCDF dataset** with two spatial dimensions and one temporal dimension (2D space × time) **.**

The dataset may contain **sea surface temperature (SST)**, **bottom temperature**, or **mean temperature for a specified water layer**. The dataset may also include a **mask defining the area of interest**.

Next, an **MHW category subset** is created, containing values from **0 to 4**, where 0 represents no MHW and values 1–4 represent the corresponding MHW intensity categories.



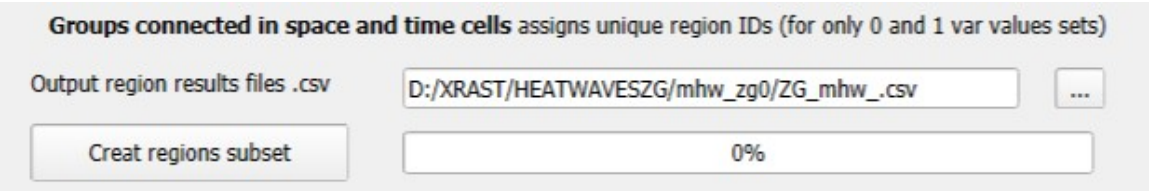
The climatology is calculated from the temperature dataset for the selected reference period (e.g., 1993–2024), together with the 90th-percentile threshold .

These two variables are then used to detect MHWs according to the **standard MHW definition**. The resulting subset contains values **1, 2, 3, and 4**, corresponding to the respective MHW intensity categories. Areas and periods where no MHW is detected are assigned a value of **0**.

The resulting MHW category subset can be further analyzed and saved as a **NetCDF dataset**.

Creation of MHW objects

The next step is to identify and create **individual MHW objects** . Each object represents a spatially and temporally coherent MHW and is assigned a unique **MHW object ID**.

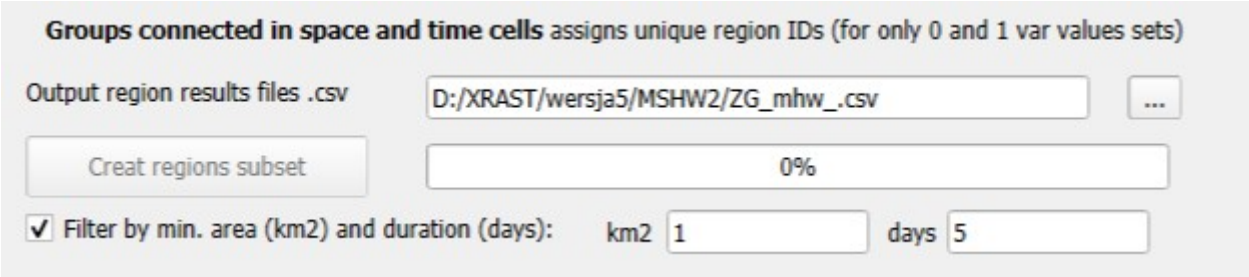


The connected cells are then **grouped according to their spatial and temporal connectivity**, and each group is assigned a unique **integer ID**. Each ID represents an individual MHW object.

The results are stored as a new subset and an accompanying **text file describing each MHW object**. This subset contains all identified objects, regardless of their duration or area. The resulting subset can be further analyzed and saved as a NetCDF dataset.

Creation of MHW objects mask

During group creation, an additional text file is generated containing only objects that meet the minimum duration and area criteria.



This file is then used to create a mask from the subset containing all identified objects, which is saved as a NetCDF file.

Final MHW Dataset

After the required variables have been calculated, a new dataset is created containing the following variables:

- **temperature**
- **day-of-year anomaly**
- **marine heatwave category**
- **MHW object ID**
- **MHW object mask**

An accompanying **text file** contains a description of the identified MHW objects and their characteristics.

The **marine heatwave category** and **MHW object ID** variables are derived solely from the temperature data and the corresponding climatology and threshold. At this stage, every detected MHW object is identified regardless of its duration or spatial extent.

The classification of an object as a **significant MHW** is determined by additional criteria, primarily its **duration and spatial extent (area)**. These criteria are applied using the **MHW object mask**, which contains binary values (**0/1**):

- **0** – the MHW object does not meet the specified duration and/or spatial-extent criteria;
- **1** – the MHW object meets the specified criteria and is included in the analysis.

MHW Analysis Step by Step

We will conduct an example MHW analysis for a part of the Mediterranean Sea using sea surface temperature data obtained from the Copernicus Marine Data Store.

The data were downloaded using the Python Application Programming Interface (API) with a dedicated script.

```
# Storing directory
DataPath = r'D:\XRAST\wersja5\MSHW'

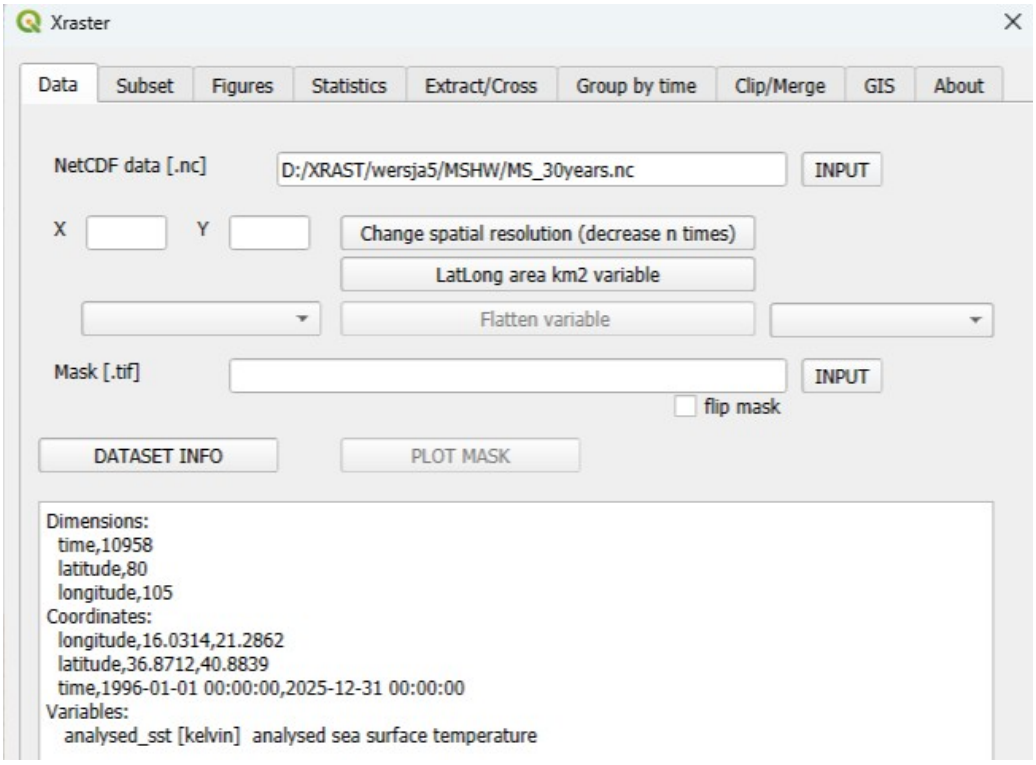
# File name
FileName = 'MS_30years.nc'

copernicusmarine.subset(
  dataset_id="cmems_SST_MED_SST_L4_REP_OBSERVATIONS_010_021",
  variables=["analysed_sst"],
  minimum_longitude=16.04494268732859,
  maximum_longitude=21.29678673772561,
  minimum_latitude=36.867424180605745,
  maximum_latitude=40.87541042959295,
  start_datetime="1996-01-01T00:00:00",
  end_datetime="2025-12-31T00:00:00",
  output_filename = FileName,
  output_directory = DataPath
)
```

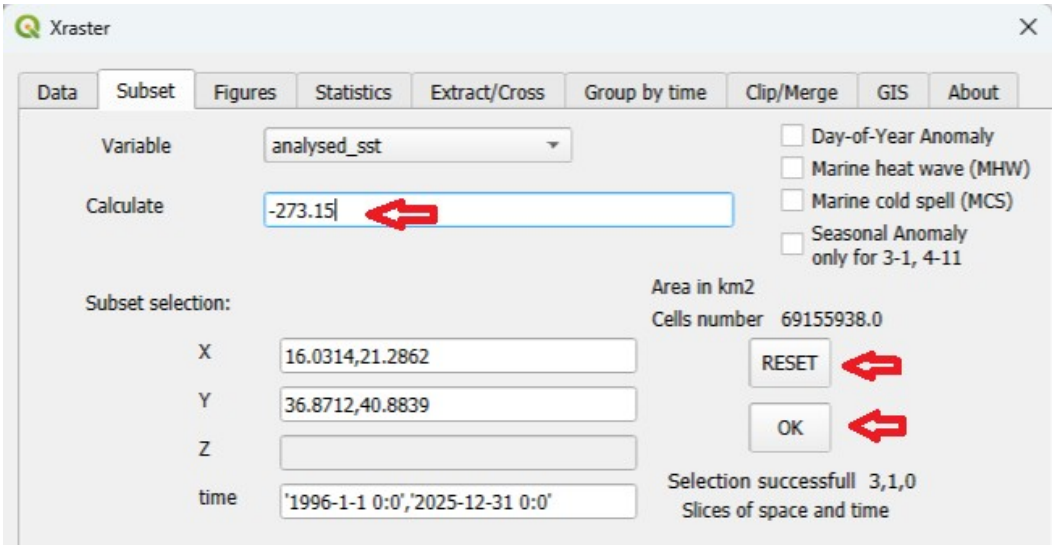
The **MS_30years.nc (180 MB) dataset can be downloaded using link**:

https://drive.google.com/file/d/1wJIQJp8bde9Ff8euJLRr5pT4L8CK3Y9Y/view?usp=sharing

We begin by **importing the data into Xraster** and becoming familiar with the dataset through **exploratory data analysis (EDA)**.

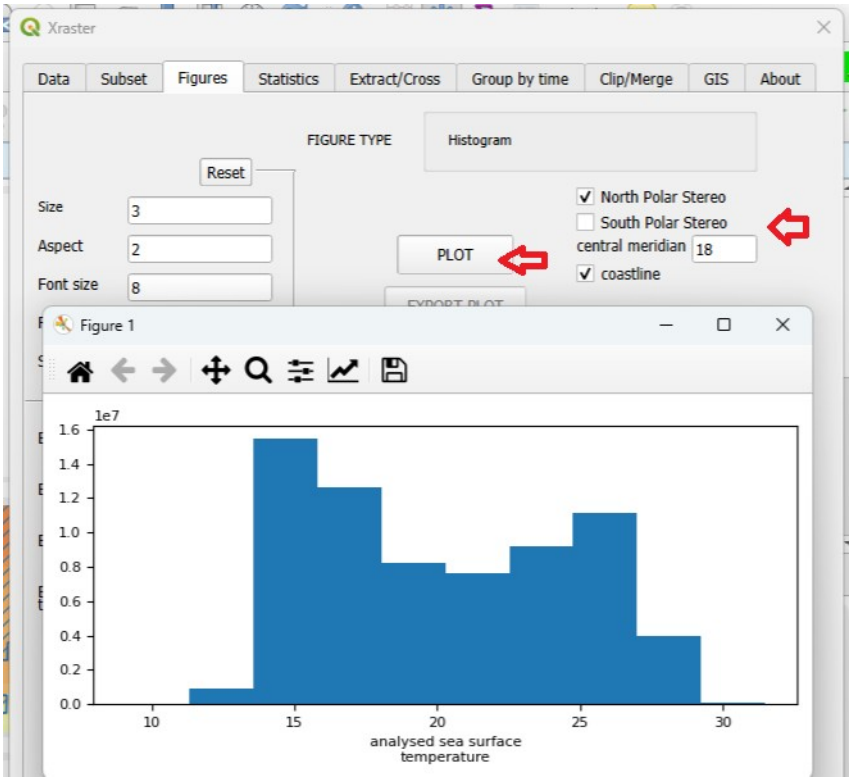


We create a **SUBSET for temperature** and convert the temperature units to **degrees Celsius**.

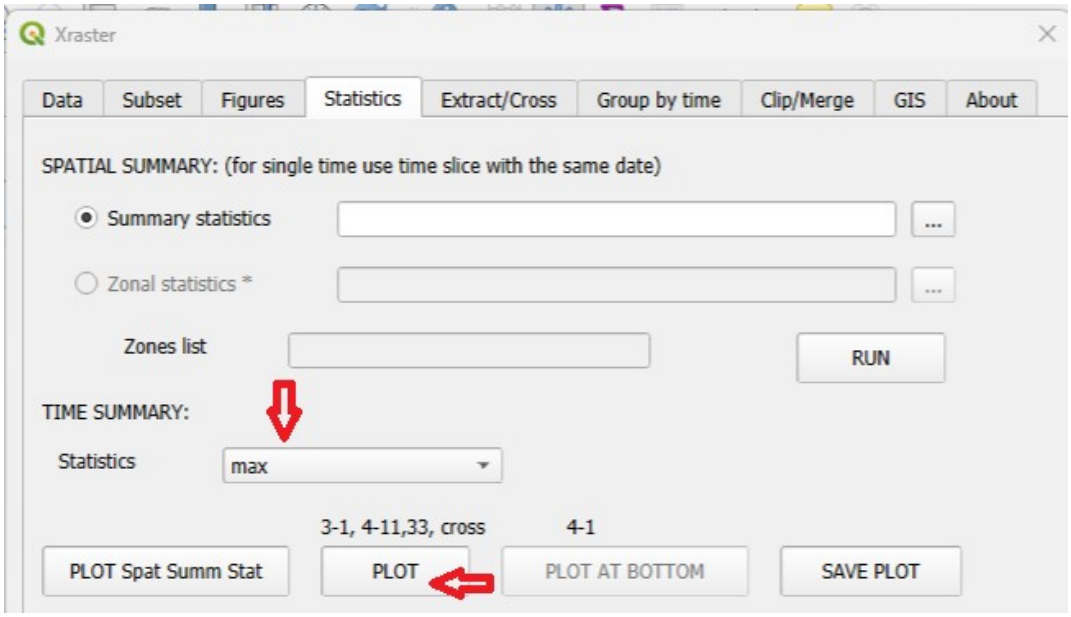


We can now create a **temperature histogram** to examine the temperature range.

Additionally, we specify the **map projection parameters** and the settings for **drawing the coastline**.

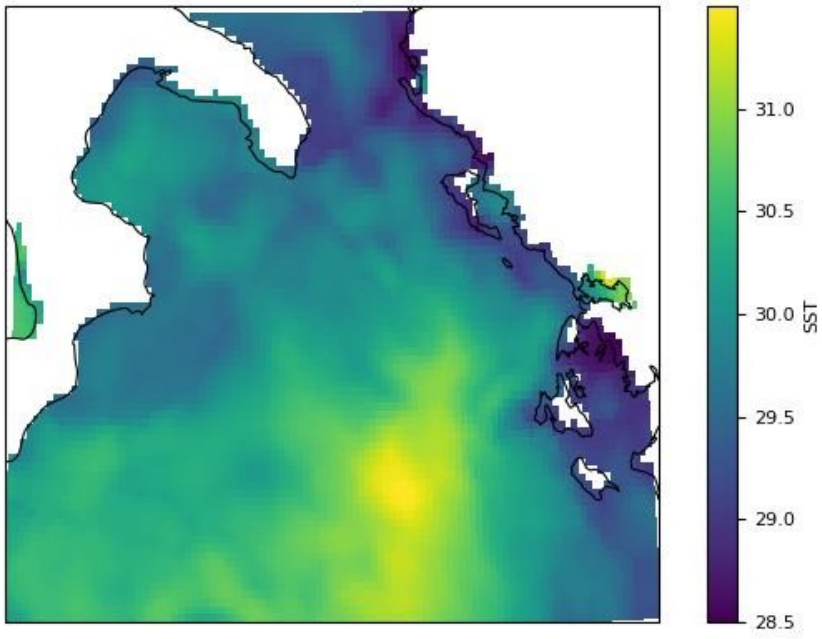


We will create a **map of the maximum temperature values**.

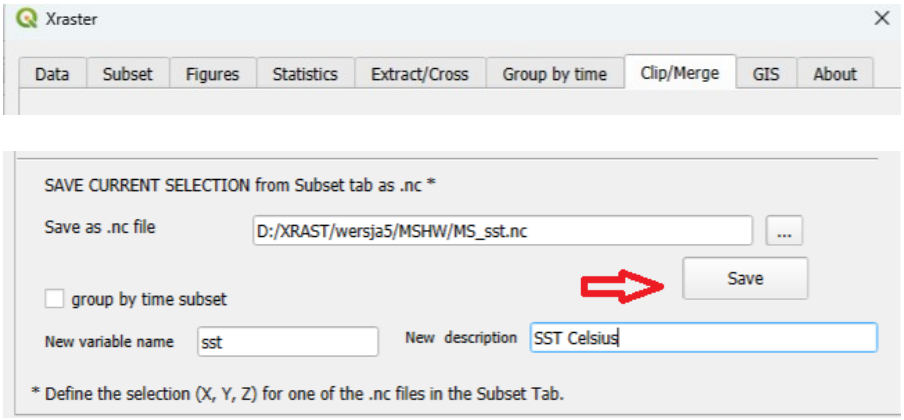


Creating the map we also visualize the **analysis area**, covering the northern part of the Ionian Sea and the coasts of Italy, Albania, and Greece, using the **maximum temperature values**.

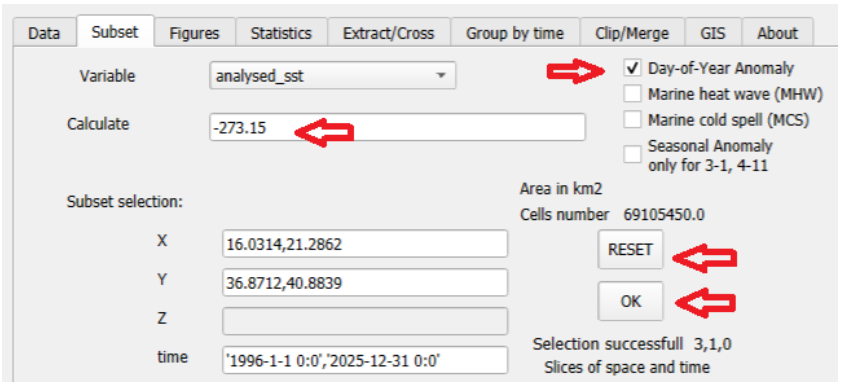
The map was created using the **save (floppy disk) icon** on the figure.



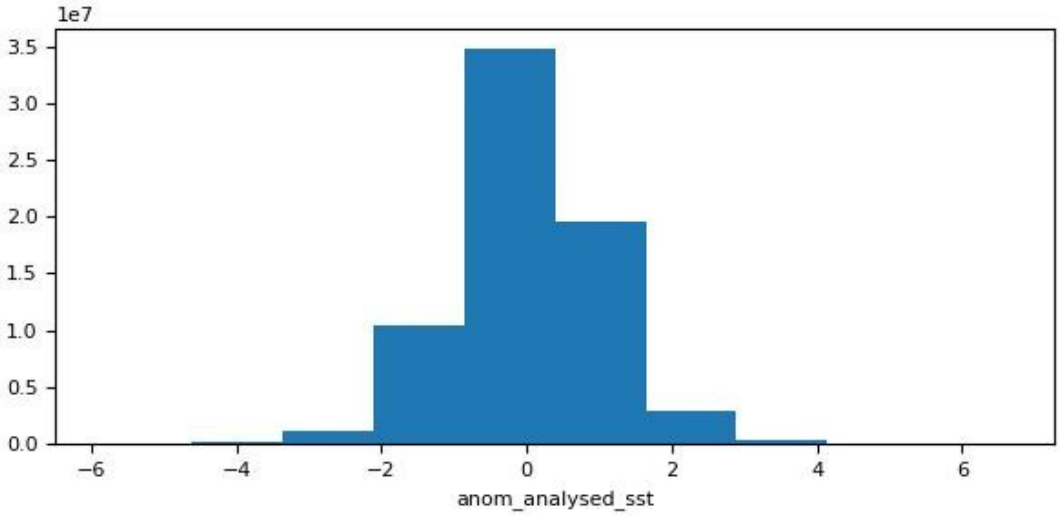
The created **SUBSET** is saved as a new **NetCDF file**.

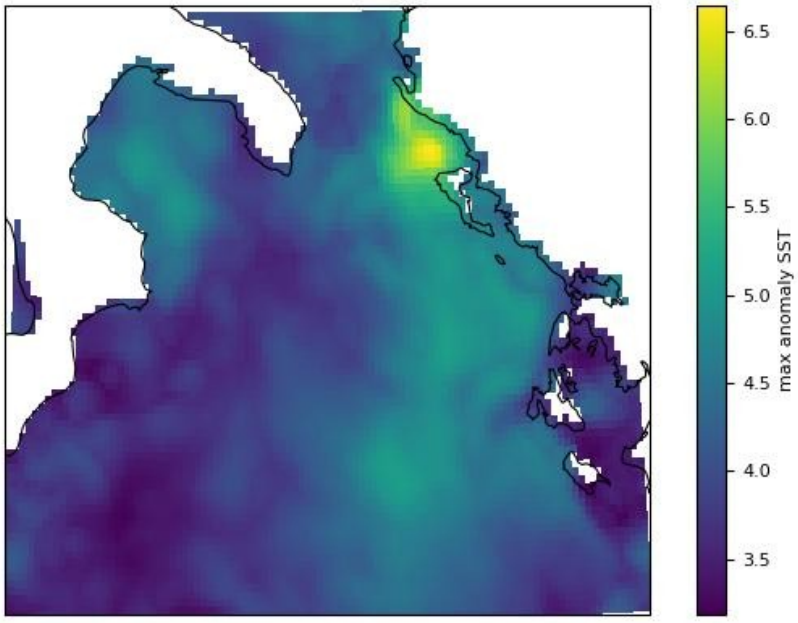


We create a **SUBSET of the Day-of-Year Anomaly**.

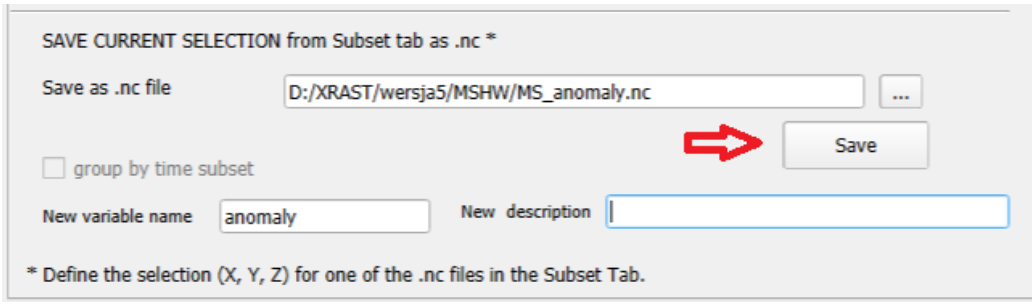


As before, we create a **histogram** to examine the range of anomaly values and a **map of the maximum anomalies**.

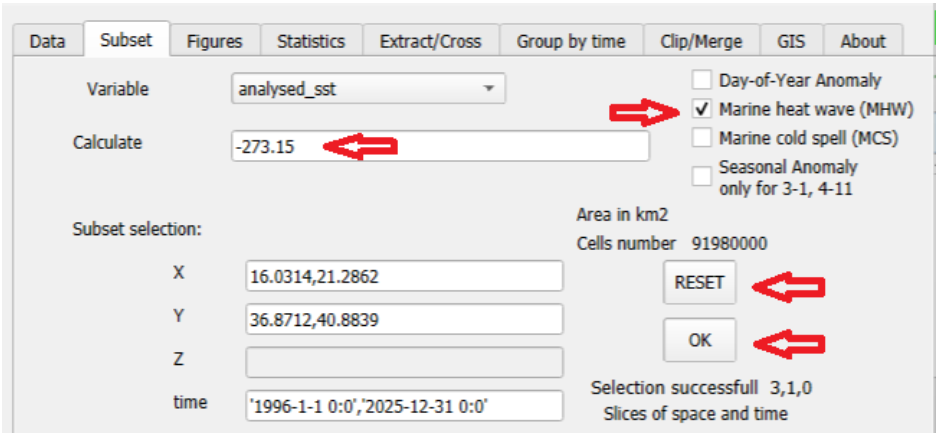




As before,
we save the created **SUBSET** as a new **NetCDF file**.



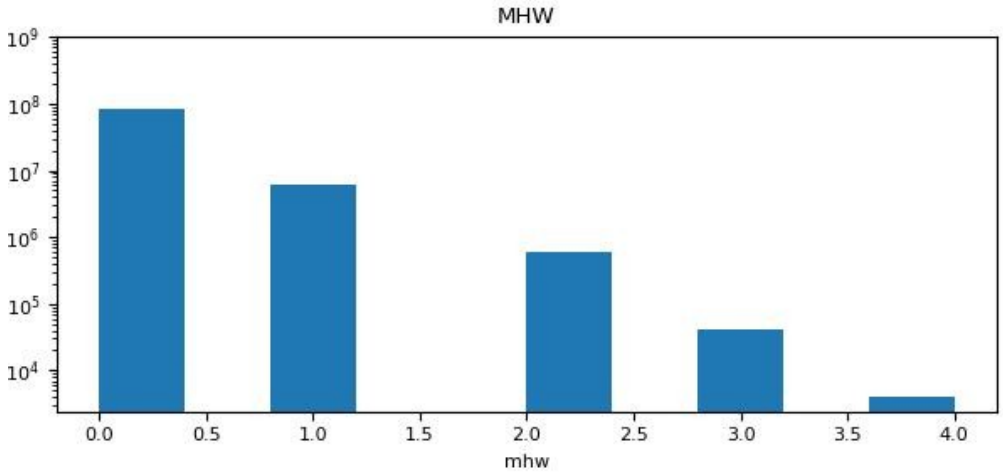
We create a **SUBSET Marine Heat Wave**. It contains four MHW categories (1, 2, 3, and 4), while 0 indicates no MHW. The categories refer to temperature anomalies, in accordance with the definition of Hobday et al. (2016), but do not include the requirement that the event lasts for at least 5 days.



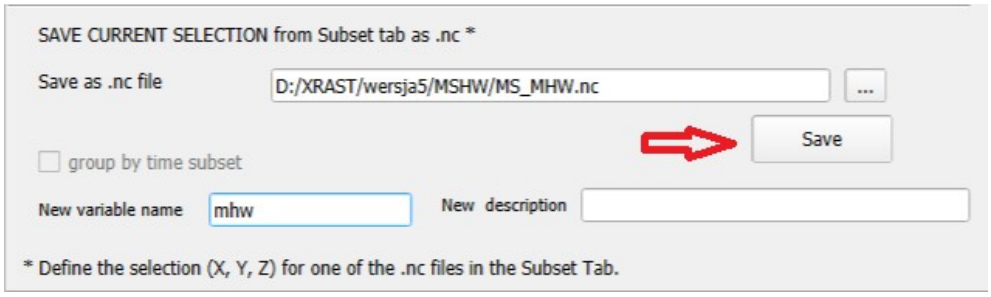
Note! This operation may take several minutes to complete.

We will create a histogram to check the resulting values (a logarithmic scale is used on the Y-axis). Click the second-to-last icon 

and select a logarithmic scale for the X-axis. You should also increase **Top** to **1e9**.

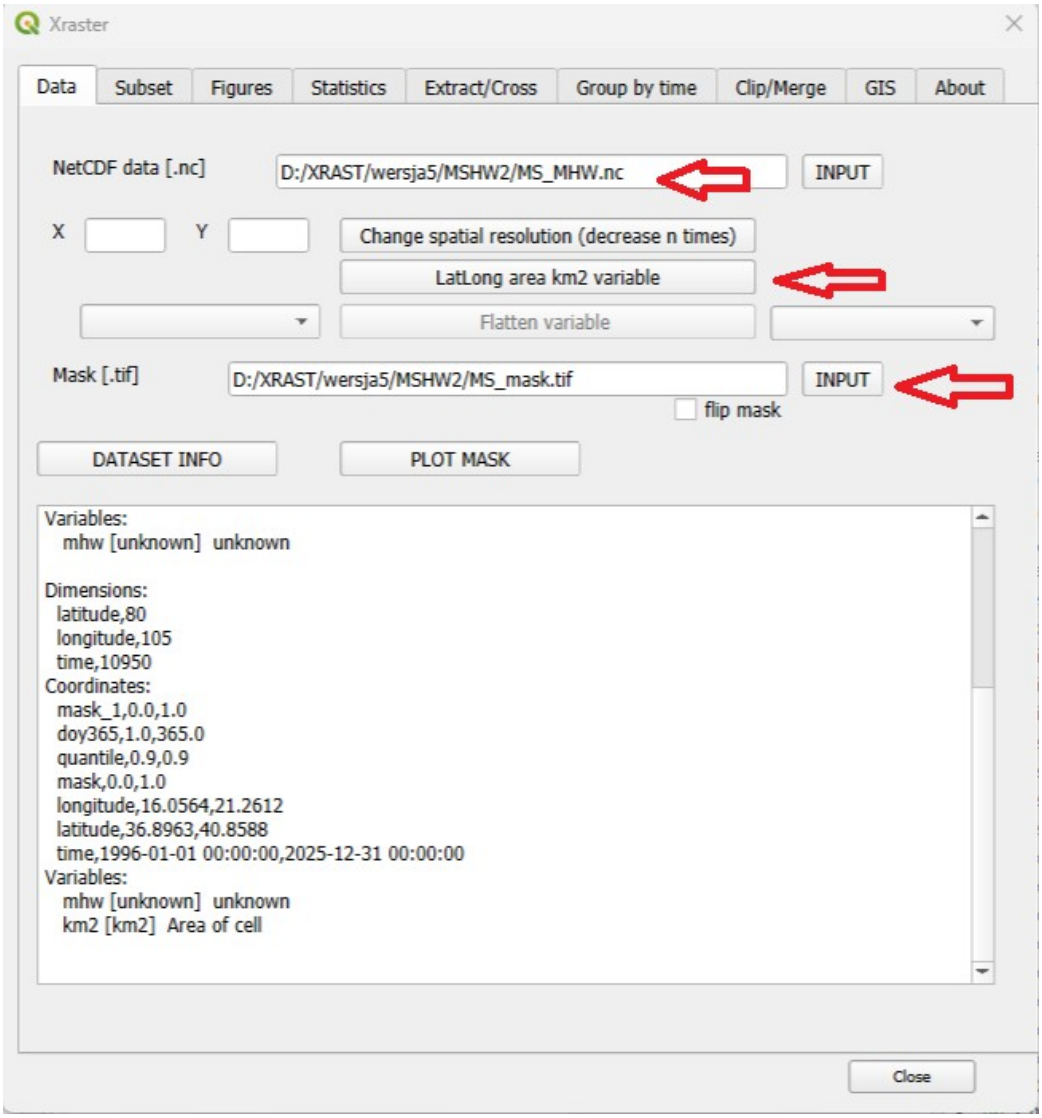


We save the completed **SUBSET** as an `.nc` file.



We will now create a **SUBSET** for the MHW objects ID. We will convert the values 1, 2, 3, and 4 to 1 using a raster algebra expression. Next, we will perform spatiotemporal grouping, assigning unique identifiers to the identified groups.

First, we will open the previously created `MS_MHW.nc` file. We will then add the mask and the grid-cell area variable.



WARNING!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

The rasters saved in `.nc` files should have identical dimensions (in this example, `latitude=80`, `longitude=105`).

The `MS_30years.nc` file has the following coordinate ranges:

- longitude = 16.0314, 21.2862
- latitude = 36.8712, 40.8839

When saving the file, these values may change slightly because the coordinates of the cell centres or edges are used.

The `MS_MHW.nc` file has the following coordinates:

- longitude = 16.0562, 21.2614
- latitude = 36.8772, 40.8779

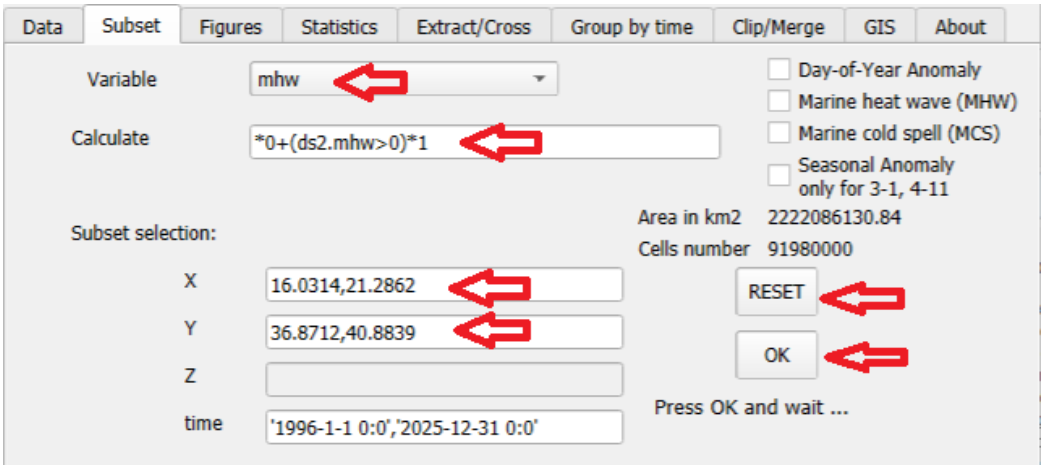
Further reduction of the coordinate range will change the dimensions of the raster.

If a new file (e.g. `MS_object.nc`) is to be created based on the dimensions of an existing file, the coordinates must be manually reset to the original values:

- longitude = 16.0314, 21.2862
- latitude = 36.8712, 40.8839

This will ensure that the raster dimensions remain unchanged.

In Subset tab we use an expression to convert mhw values to 1.



The X and Y values have been updated.

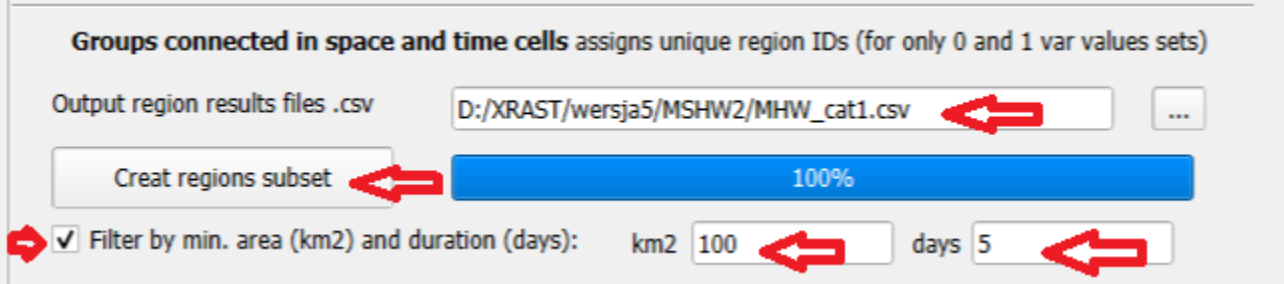
The expression is:

```
mhw * 0 + (mhw > 0) * 1
```

(In the **Calculate** field, we enter the part of the expression following the variable name.)

If **mhw** is greater than zero (logical value = 1), the result is 1; otherwise, the result is 0.

We open the **GIS** tab, enter the name of the text file containing the results of the operation, and click the **Create regions subset** button. Additionally, we will filter the identified objects based on their area and duration.



The progress of the calculations can be monitored on the progress bar, which shows the percentage completed.

A new **SUBSET** is created containing the object IDs (**region ID**) and three text files:

MHW_cat1.csv,
MHW_cat1_sum.csv

MHW_cat1_sumF.csv.

The **MHW_cat1_sum.csv** file contains information about the identified objects, sorted by their maximum area. It has the following form:

id,max_number_of_cells,date_start,date_end,date_of_max
38729,6268,2023-07-11,2023-08-07,2023-07-24
27307,6268,2018-04-14,2018-06-21,2018-05-30
39981,6268,2023-12-21,2024-05-03,2024-04-08
42244,6268,2024-07-08,2025-04-18,2024-08-18
7118,6268,2002-03-18,2002-03-26,2002-03-22
5124,6268,2001-01-25,2001-04-13,2001-03-25
35853,6263,2022-06-19,2022-07-12,2022-07-03
18675,6261,2013-04-24,2013-05-14,2013-05-03
22868,6255,2015-08-26,2015-10-16,2015-09-20

.....

This file contains more than 14,000 objects, but because they are sorted by maximum area (**max_number_of_cells**), we can easily identify the most significant objects in terms of area. We also have information about when each object started and ended, as well as the date on which it reached its maximum area.

The **MHW_cat1_sumF.csv** file contains 671 objects remaining after filtering based on their area and duration. These objects meet the MHW criteria, including the minimum duration of 5 days. Additionally, the file contains information on the area (km²) and duration (days) of each object.

id,max_number_of_cells,date_start,date_end,date_of_max,days,area_km2

38729,6268,2023-07-11,2023-08-07,2023-07-24,28,424.62151142944336

27307,6268,2018-04-14,2018-06-21,2018-05-30,69,151424.62142944336

39981,6268,2023-12-21,2024-05-03,2024-04-08,135,151424.62142944336

42244,6268,2024-07-08,2025-04-18,2024-08-18,285,151424.62142944336

7118,6268,2002-03-18,2002-03-26,2002-03-22,9,151424.62142944336

5124,6268,2001-01-25,2001-04-13,2001-03-25,79,151424.62142944336

35853,6263,2022-06-19,2022-07-12,2022-07-03,24,151303.82961273193

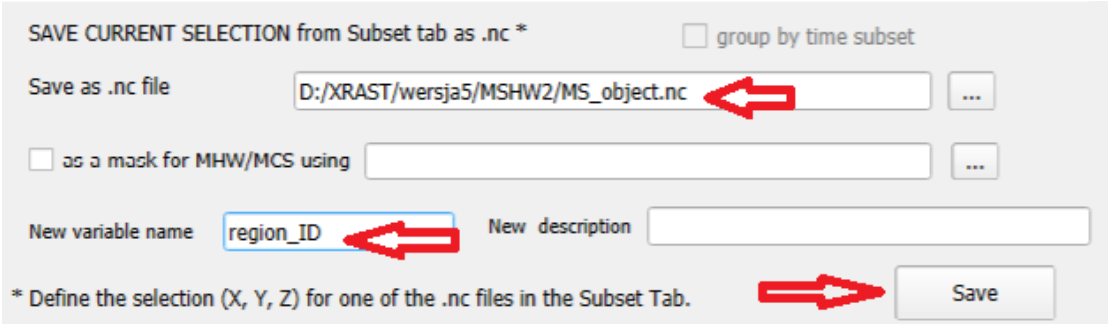
18675,6261,2013-04-24,2013-05-14,2013-05-03,21,151255.51288604736

22868,6255,2015-08-26,2015-10-16,2015-09-20,52,151110.56270599365

.....

We can now analyse individual objects using the tools available in this panel.

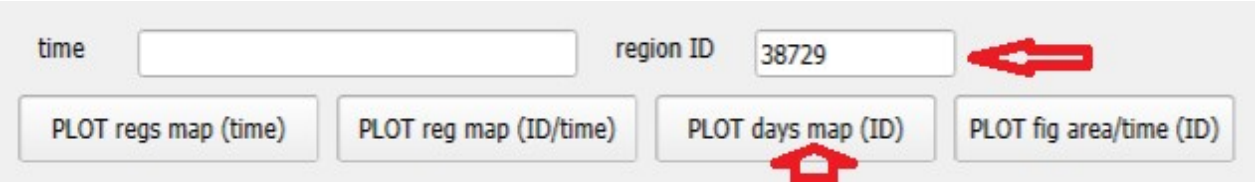
First, however, we will save it as a fourth file containing the **region_ID** variable.



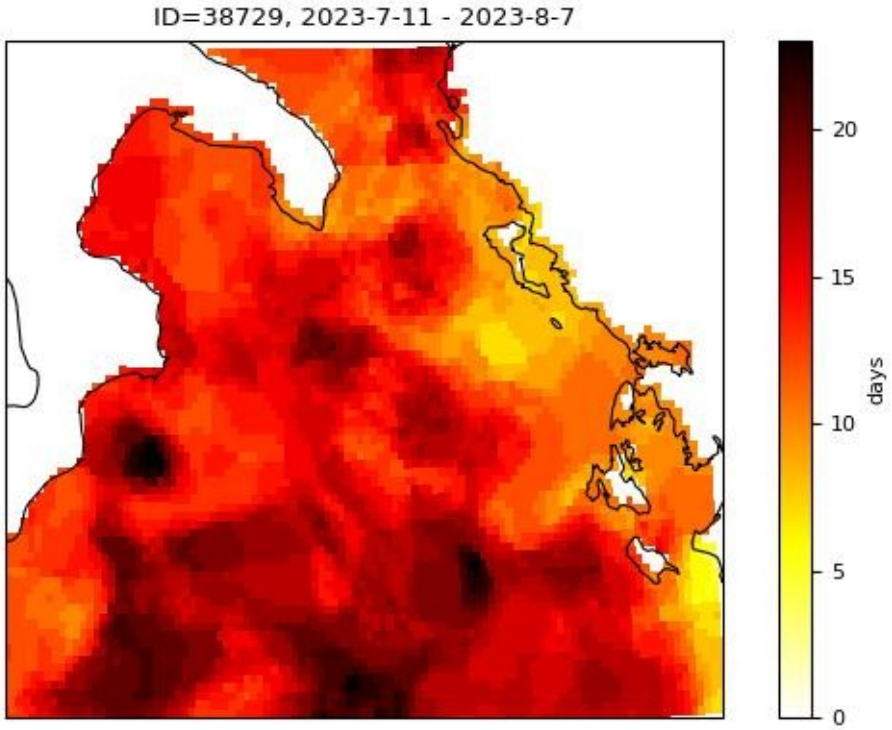
Since the region ID subset remains in memory after completing the procedure, we can perform a series of analyses using the tools available in this tab.

We will analyse the first object, with **ID = 38729**.

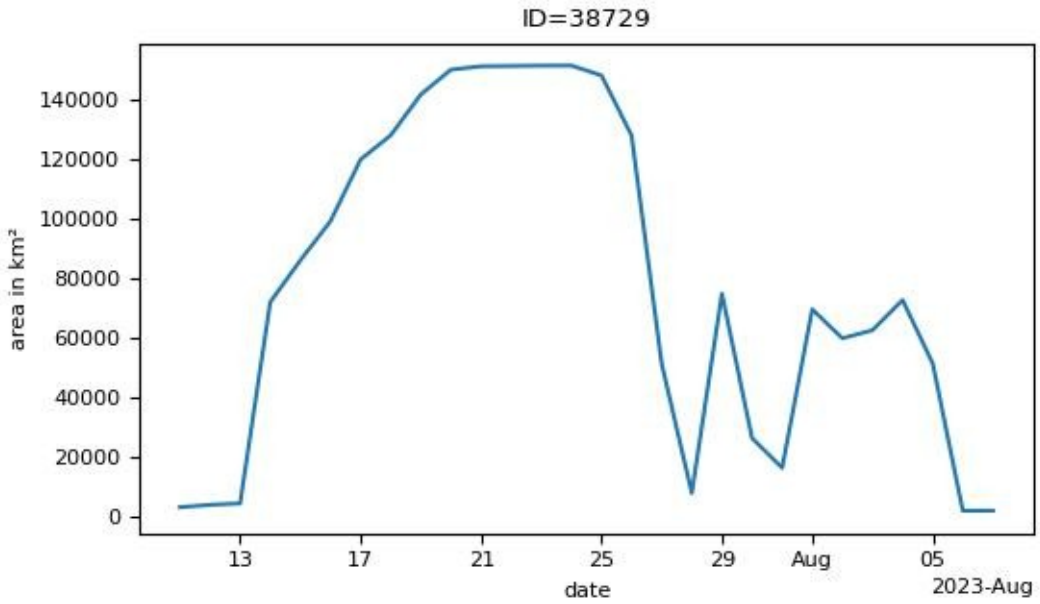
The tool provides four buttons that enable the visualization of objects according to time and their ID.



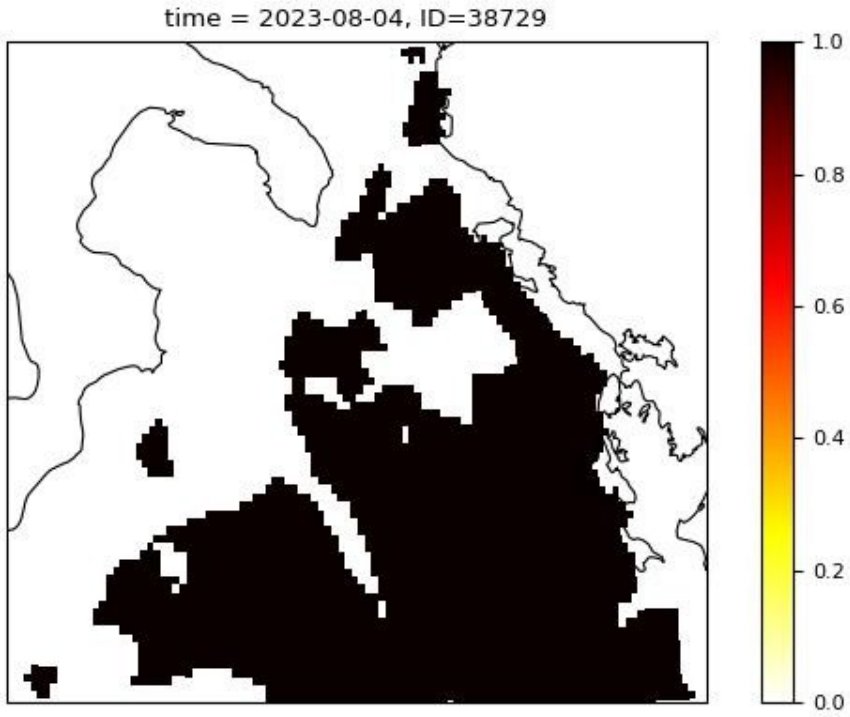
Entering the **region ID** and clicking the **PLOT days map** button produces a map showing the number of days for which the given MHW (ID = 38729) existed in each cell.



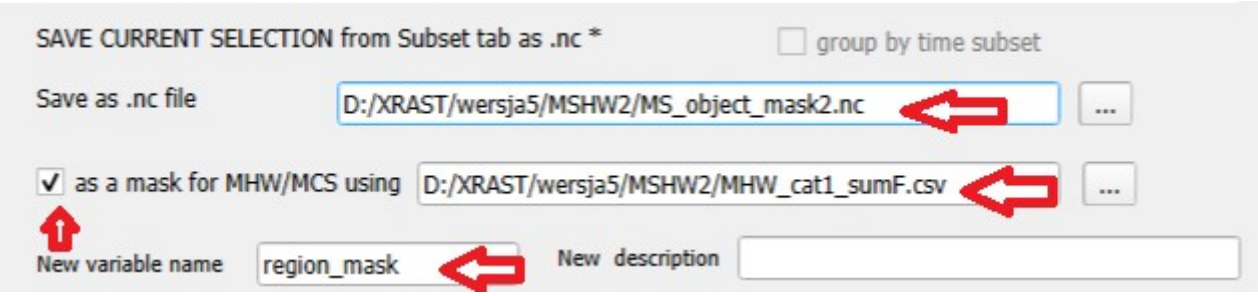
Entering the **region ID** and clicking the **PLOT fig area/time** button produces a plot showing changes in the area of the given MHW (ID =38729) over time.



Entering the ID and date (e.g. 2023-08-04) allows us to obtain the location of the MHW object on a given day by clicking the **PLOT reg map (ID/time)** button.



In the final step, we will create a **SUBSET** for the MHW objects mask. The mask will contain a value of 1 where MHWs occur and 0 elsewhere. To create it, we will use the **SUBSET** containing the group IDs currently in memory and the filtered text file.



As a result, we currently have the following files:

NetCDF files:

- **MS_sst.nc** — variable: **sst** (in degrees Celsius)
- **MS_anomaly.nc** — variable: **anomaly** (day-of-year SST anomaly)
- **MS_MHW.nc** — variable: **mhw** (categories 1, 2, 3, 4; 0 indicates no MHW)
- **MS_object.nc** — variable: **region_ID** (region object ID)
- **MS_object_mask2.nc** - variable: **obj_mask** (MHW binary mask)

Text files:

- **MHW_cat1_sumF.csv**

We will now combine the .nc files: into a single file:

MS_sst_anom_mhw_obj_mask.nc

containing all five variables.

We create a text file (**nc_files.csv**) containing the file paths.

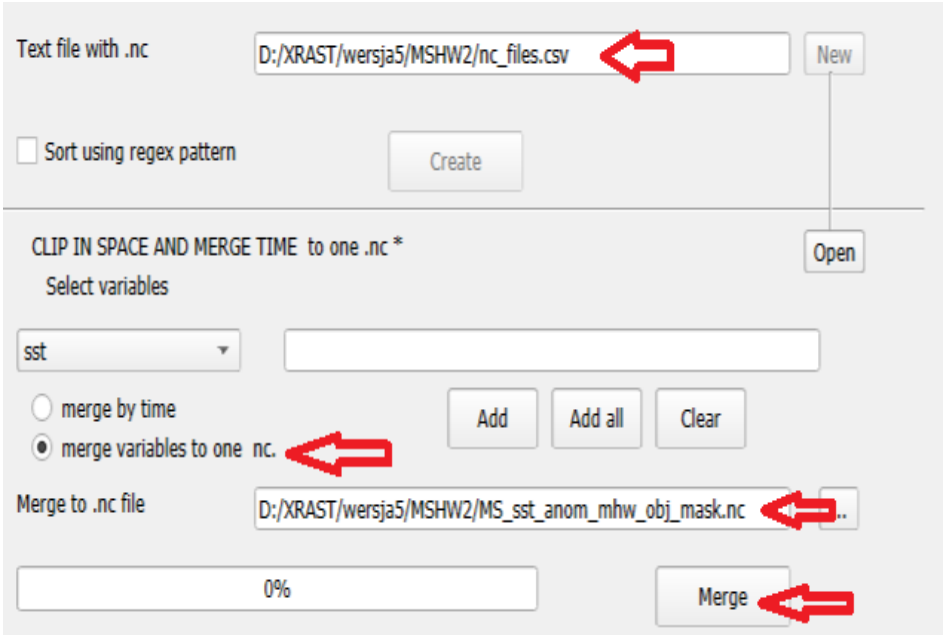
D:\XRAST\wersja5\MSHW2\MS_sst.nc
D:\XRAST\wersja5\MSHW2\MS_anomaly.nc
D:\XRAST\wersja5\MSHW2\MS_MHW.nc
D:\XRAST\wersja5\MSHW2\MS_object.nc
D:\XRAST\wersja5\MSHW2\MS_object_mask2.nc

The operation of combining the files into a single file containing five variables is performed in the **Clip/Merge** tab. The settings should be entered as shown in the figure below.

Important:
All files must have the same number of rows and columns.

Before performing the operation, enter a representative subset (e.g. with the downloaded data) (**Subset → Reset → OK**) to define the **X, Y, and Z** dimensions of the output.

If the NetCDF file created by the program (after any operation) has too few rows or columns, usually by one, the **X** and **Y** ranges can be adjusted slightly in the **Subset** tab.



The created file

MS_sst_anom_mhw_obj_mask.nc

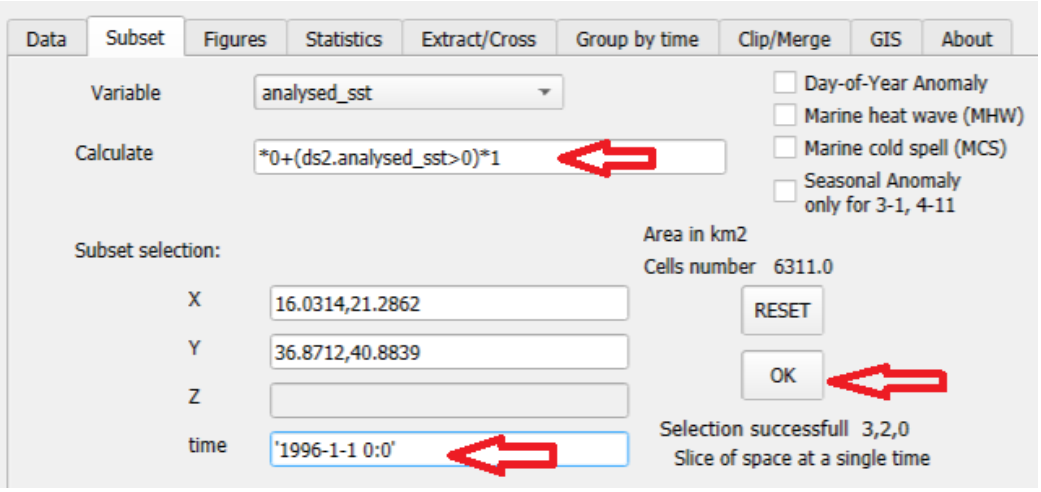
contains five variables:

Variables:

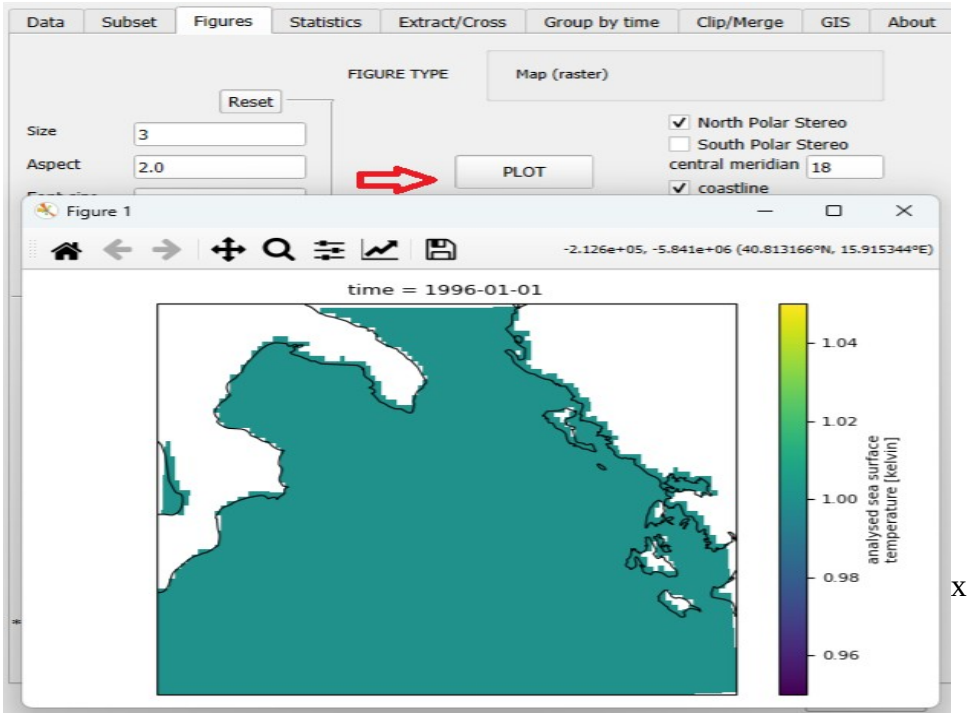
- **sst [°C] — SST Celsius**
  SST in degrees Celsius.
- **anomaly [unknown] — unknown**
  Climatic SST anomaly (relative to the mean value in each cell for a given day of the year). This means that the baseline used to calculate the anomaly is flat.
- **mhw [unknown] — unknown**
  MHW category, determined by how much the sea temperature exceeds the **90th percentile threshold** relative to the local climatology. The categories should not be considered equivalent to MHWs according to the Hobday definition because they do not include the requirement that the event persist continuously for at least 5 days.
- **region_ID [unknown] — unknown**
  ID of the MHW object (**region group**).
- **obj_mask [unknown] — MHW binary mask**
  A binary mask containing **1** when an MHW is present in a cell (according to the Hobday definition) and **0** when no MHW is present.

++

Additionally, we will create a **land–water mask** for our data. We create a single layer (for one day) containing values of **1** and **NoData** (from any available data).

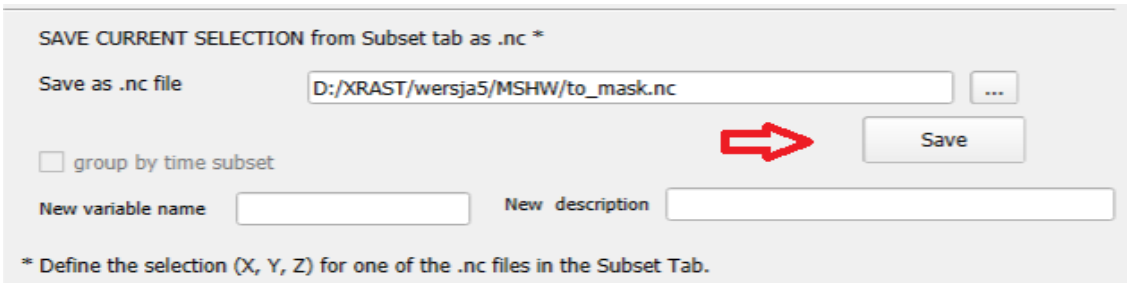


The map will have the following form:

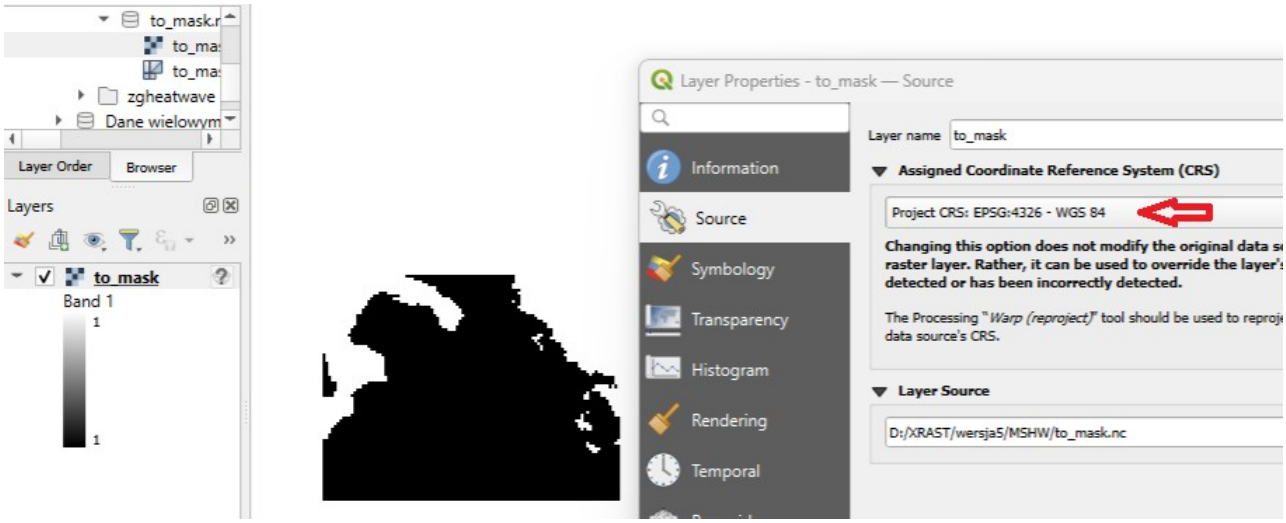


x

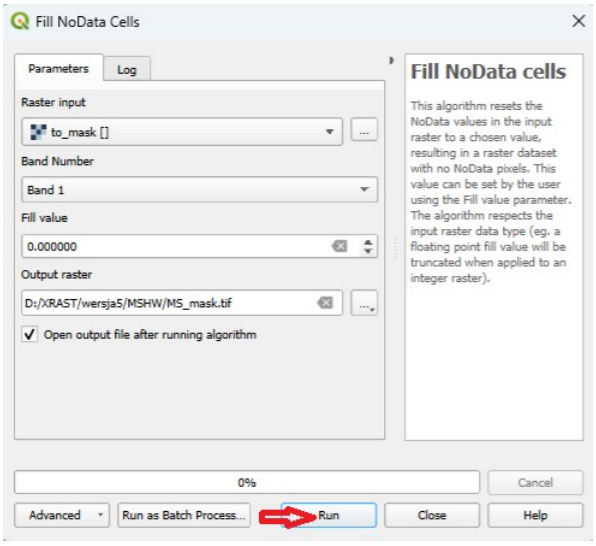
We save it as a new NetCDF file.



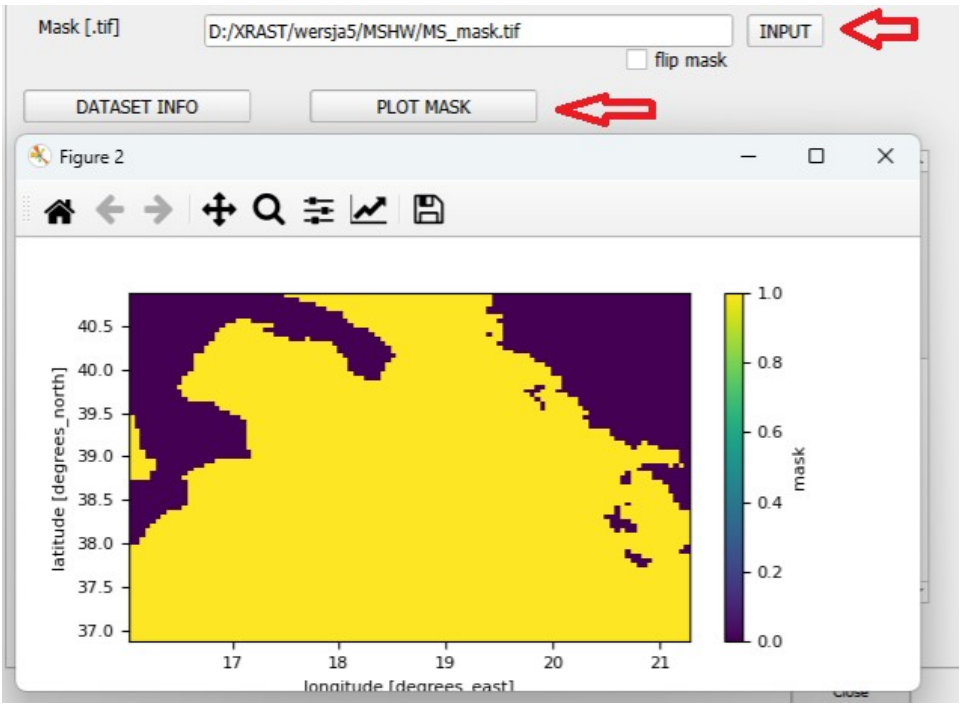
We open the file in QGIS and set the geographic coordinate reference system.



We perform the operation of filling **NoData cells** with the value **0** and save the result as a **.tif** file.



We can now use the mask in **Xraster**.



++

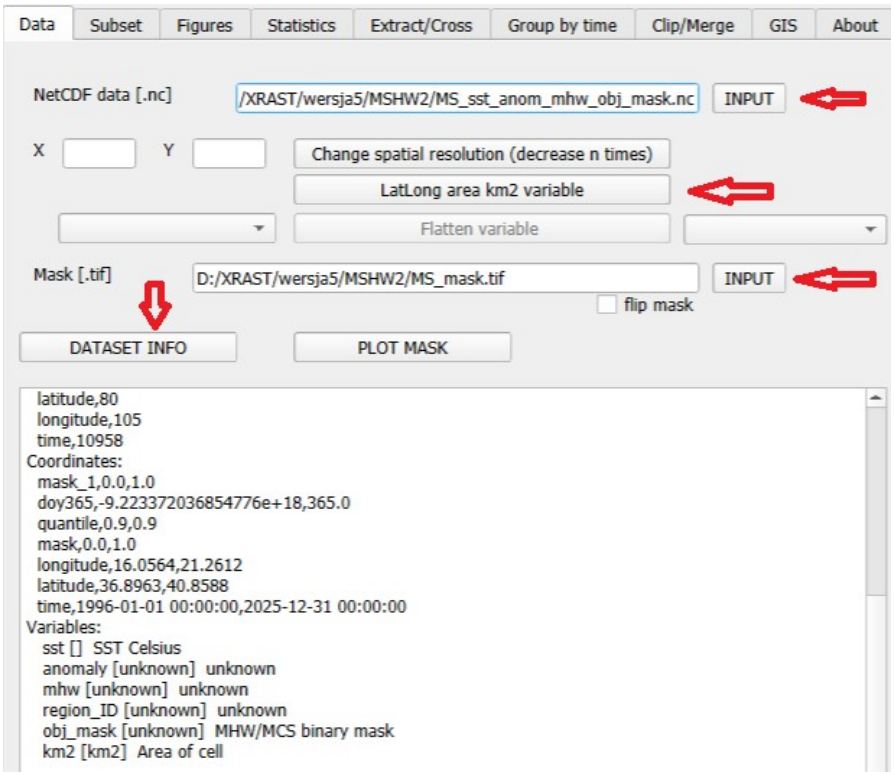
Global studies of MHWs, based on pixel values, usually include the following maps:

- **Mean annual MHW days** — map of the mean number of MHW days per year.
- **Mean frequency** - events yr^{-1}
- **Mean duration** - średnia długość trwania MHW (mean annual MHW days / mean frequancy)
- **Mean intensity** — map of the mean temperature anomaly during the MHW period.
- **Maximum intensity** — map of the maximum temperature anomaly during the MHW period.
- **Cumulative intensity** — sum of temperature anomalies during the MHW period.
- **Maximum MHW category** — map of the maximum MHW category.

They also include **time-series graphs** showing how the regional or global averages and maximum values of these metrics change over time.

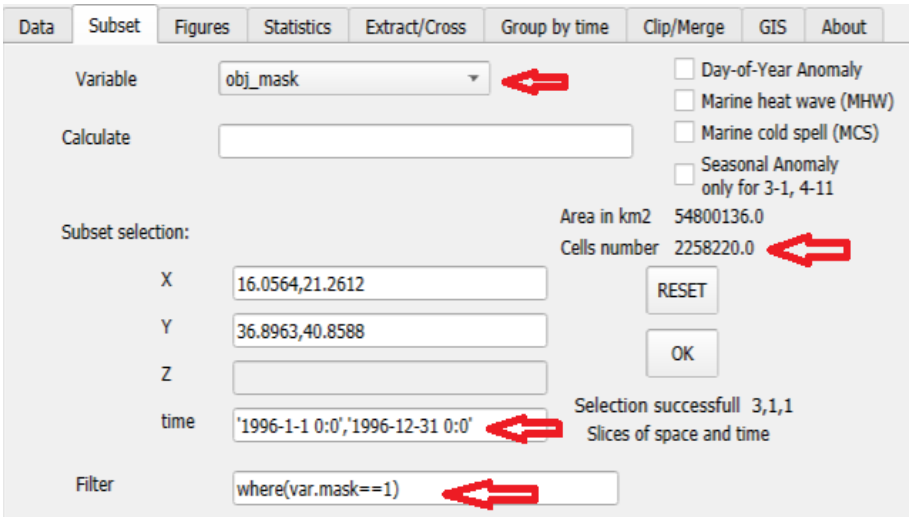
Plot of the mean number of MHW days for individual years.

In the **Data** tab, enter the data and the mask, and create the **km²** variable (using the *LatLong area km2 variable* button). After entering the data, click **DATASET INFO** to view all variables (the mask can be found under **Coordinates**).



We will use the **obj_mask** variable (1 when MHW occurs in a cell, 0 when it does not). To calculate the annual mean, we **sum the variable over the year** and divide by the number of **cells in the year**. This gives a value from 0 to 1, representing the **MHW frequency**. To convert it to days, we multiply the value by 365.

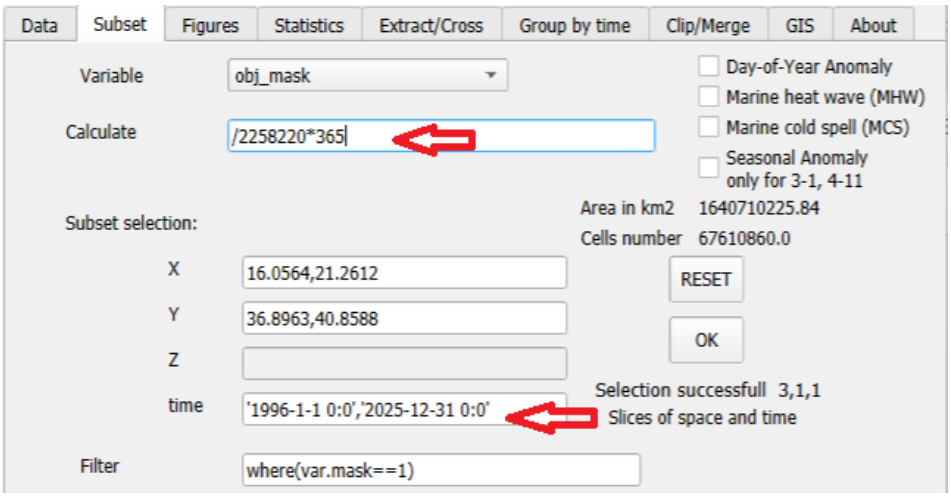
First, in the **Subset** tab, we determine the number of days per year within the area defined by the mask with a value of 1.



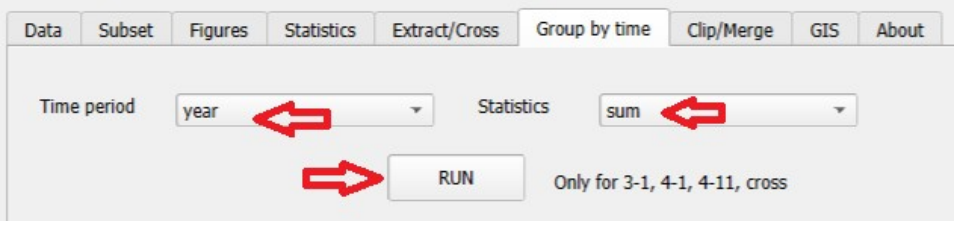
To perform the required operation, we need to assign each cell a value equal to the variable divided by the number of cells in the year (2258220) and multiplied by the number of days in the year.

For example, if there were 10 cells and half of them had a value of 1 (the remaining cells having a value of 0), five cells would receive the value **1/10 × 365**, while the other five would receive **0/10 × 365**. Their sum would give **182.5 mean days per year**.

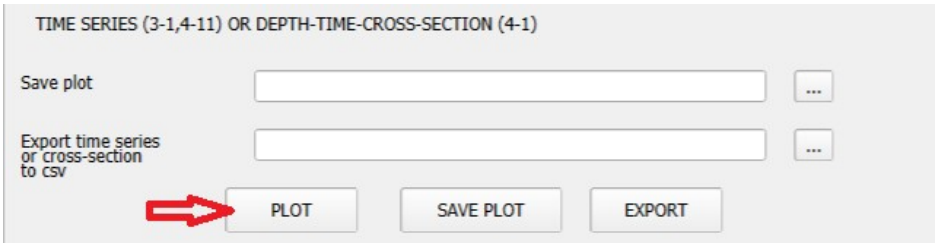
The expression in the **Calculate** field would therefore be **/2258220*365**, which is equivalent to the Map Algebra expression **variable / 2258220 * 365**.



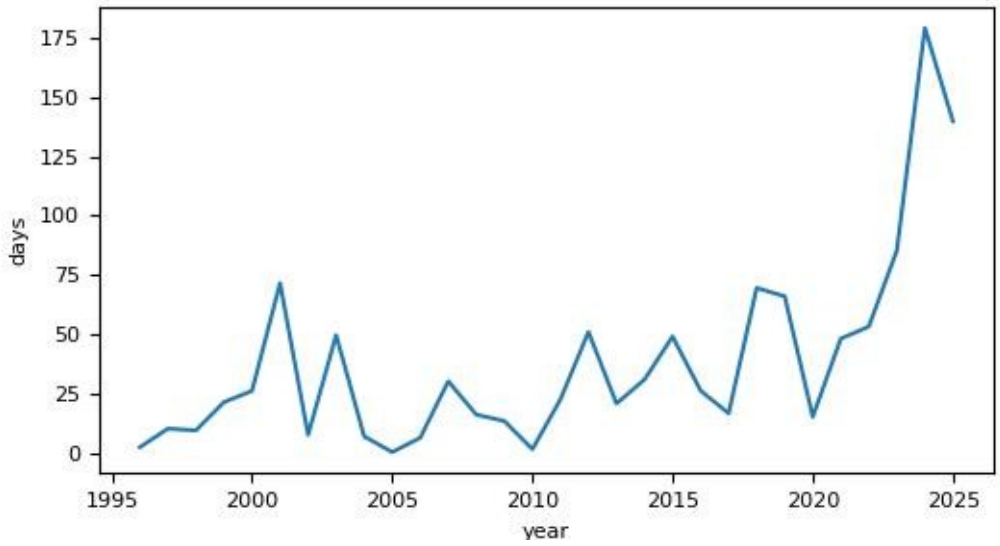
We move on to the Group by Time tab.



and click the **PLOT** button.

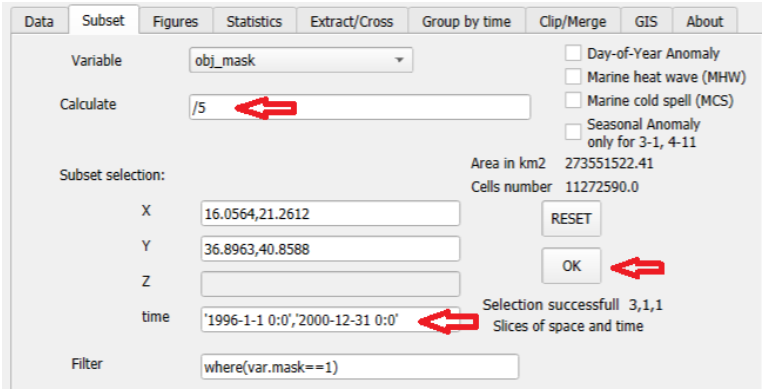


We obtain a plot showing how the number of days (spatial average) changed in individual years. The figures and maps presented have been edited using a graphics program (the editing methods are described in the manual).



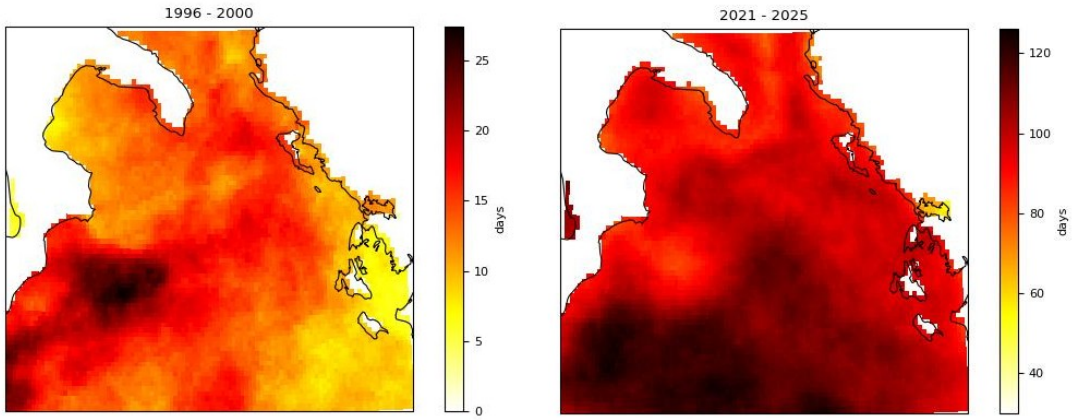
Due to the changes observed over the last three years, further mapping (mean number of MHW days, mean and maximum intensity, and cumulative intensity) will be performed for two periods: **1996–2000** and **2021–2025**. In the **Subset** tab, we create subsets of the variable divided by 5 (5 years).

Map of mean number of MHW days,



Next, in the **Statistics** tab, under the **TIME SUMMARY** panel, enter **sum** in the **Statistics** field and click **PLOT**.

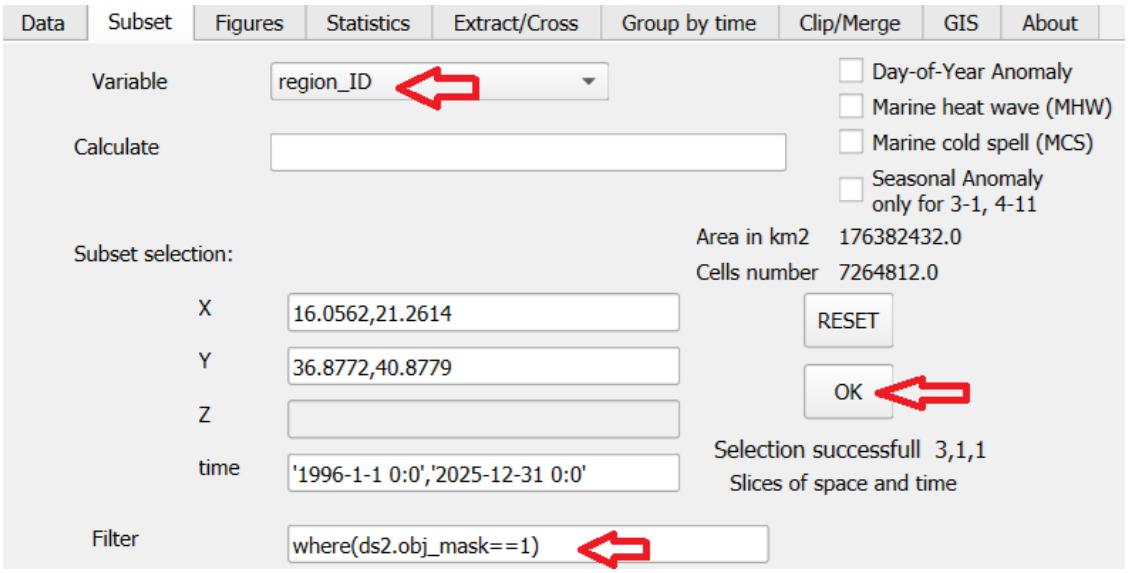
We repeat these steps for both periods. We will obtain the maps.



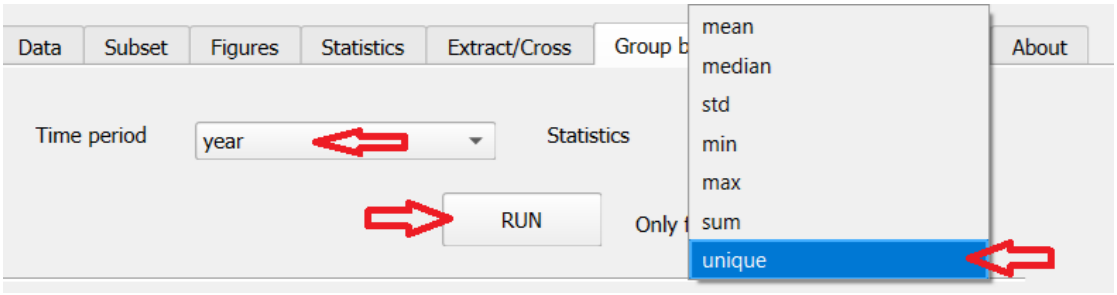
Maping of MHW frequancy

Mapping MHW frequency involves creating maps or time series showing the number of MHW occurrences over a given period. We will start by creating a plot showing how the number of MHW events changed over the individual years.

We create a subset containing the `region_ID` variable masked by `obj_mask`. Only the IDs of groups that meet the minimum duration and area criteria are retained.

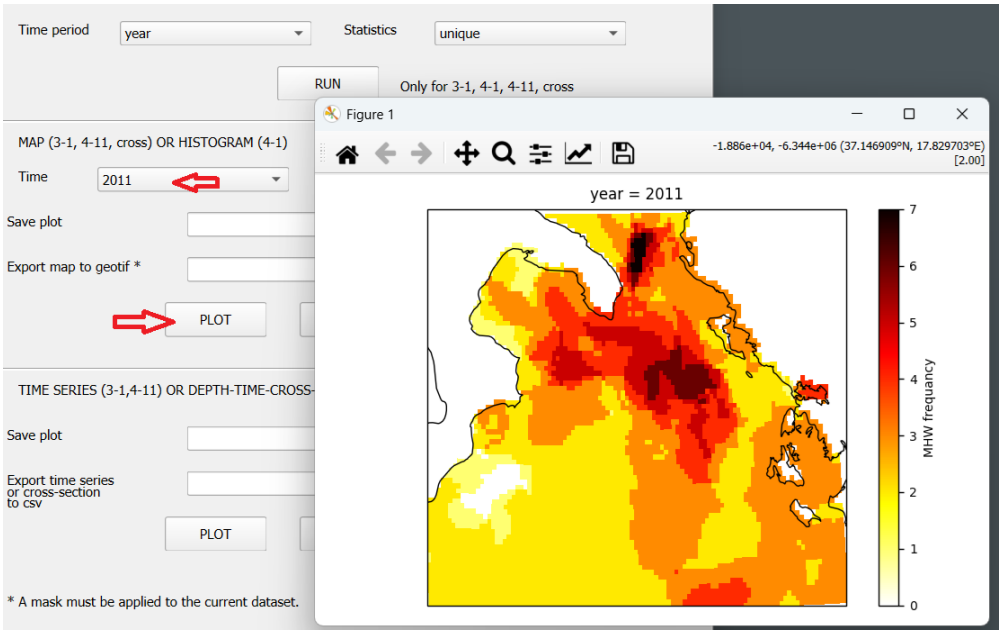


We proceed to the Group by Time tab.

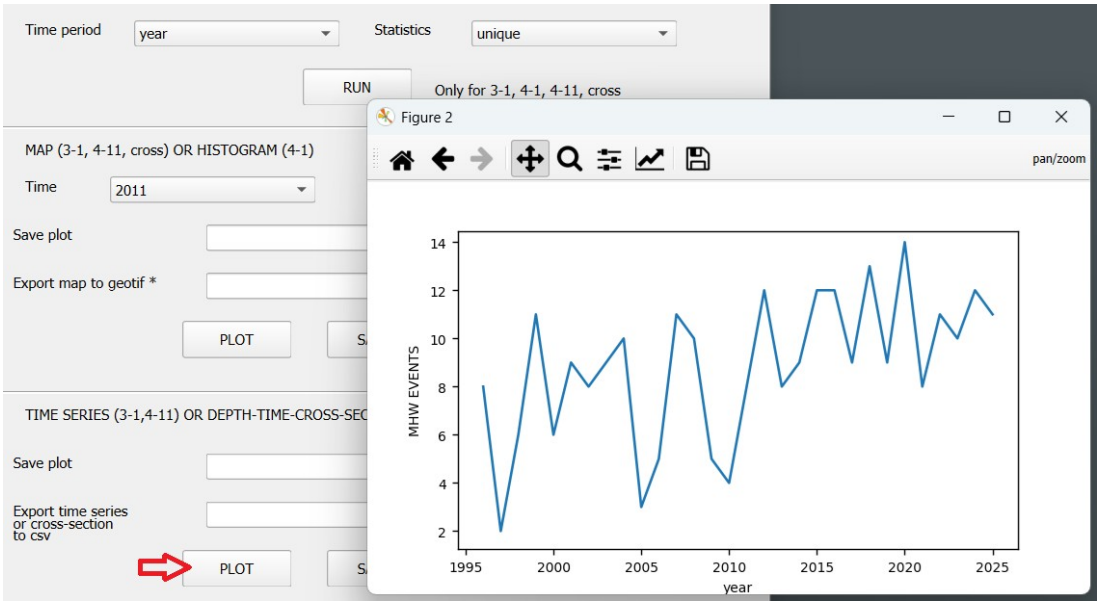


Select **year** as the time grouping interval and **unique** as the statistic. Two types of output can be generated.

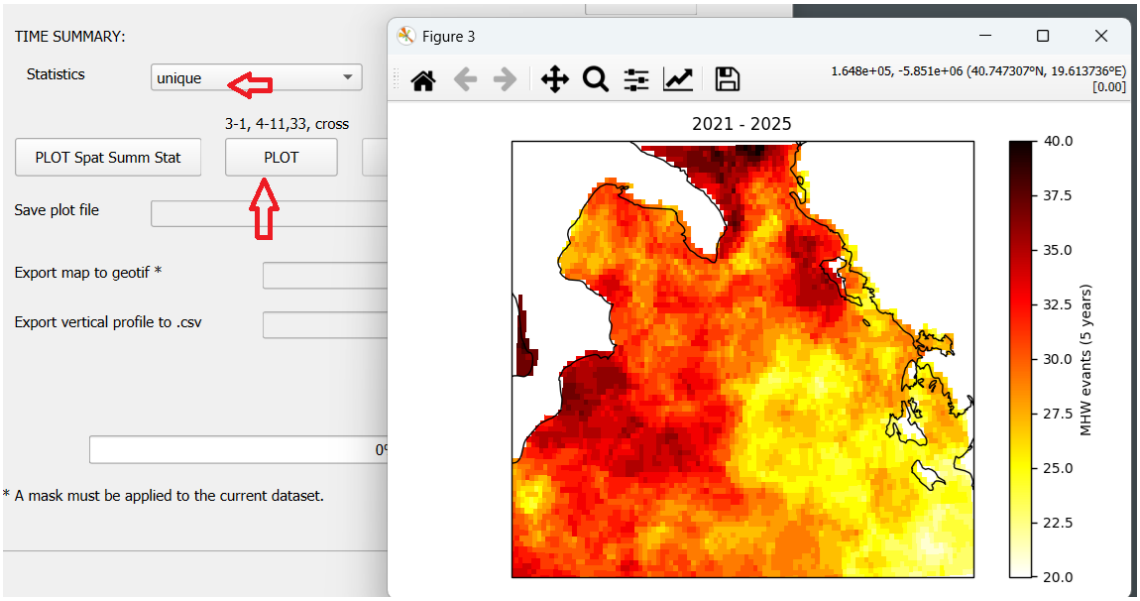
Maps for individual years, in which each grid cell is assigned the number of unique group IDs, representing the number of MHW events.



The second output is a time series showing the number of MHW events in each year.



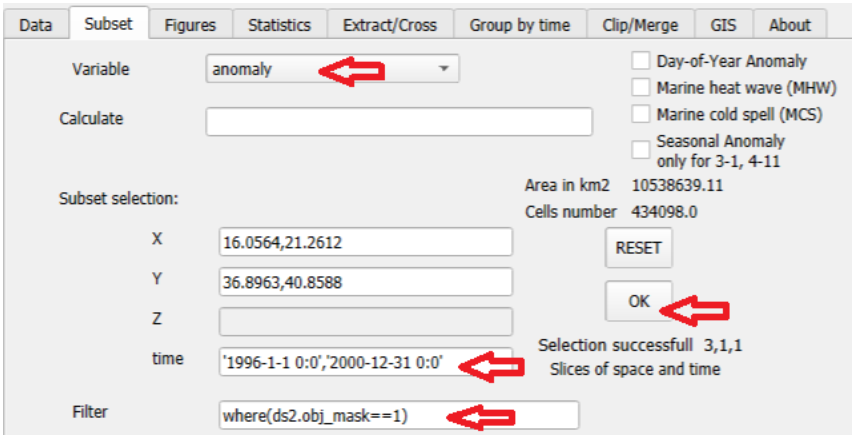
Maps for any selected periods can be generated in the Statistics tab.



This map shows the number of MHW events over a five-year period. To calculate the **mean annual MHW frequency** for this period, export the map to QGIS and divide its values by 5 using the Raster Calculator.

Map of mean intensity

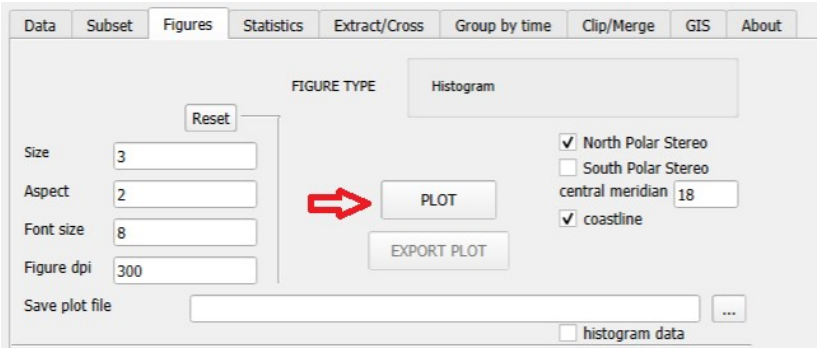
We create maps of the mean temperature anomalies during the duration of MHW events (additionally, we include histograms on the maps).



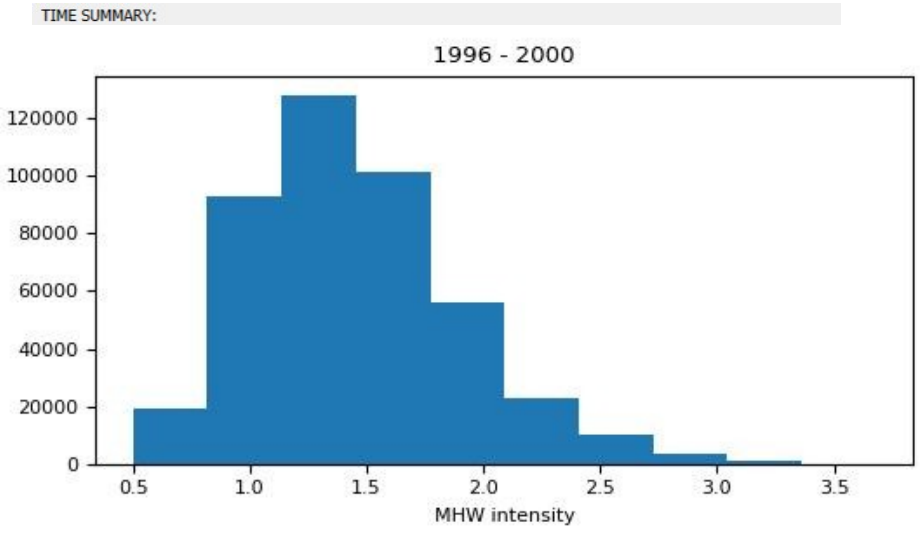
We enter the **anomaly** (intensity) variable for the first five-year period. We create a subset only for cells classified as MHW using the following filter:

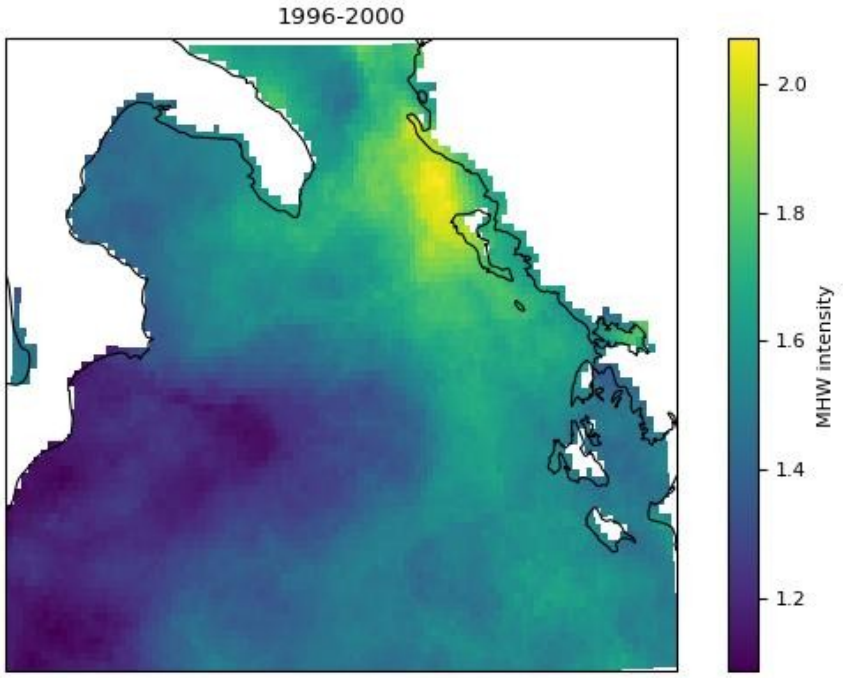
where(ds2.obj_mask==1)

The histogram is created in the **Figures** tab. The histogram data can be saved as a text file by selecting *histogram data* and saving the plot using *Save plot file*.



The map is created in the **Statistics** tab.



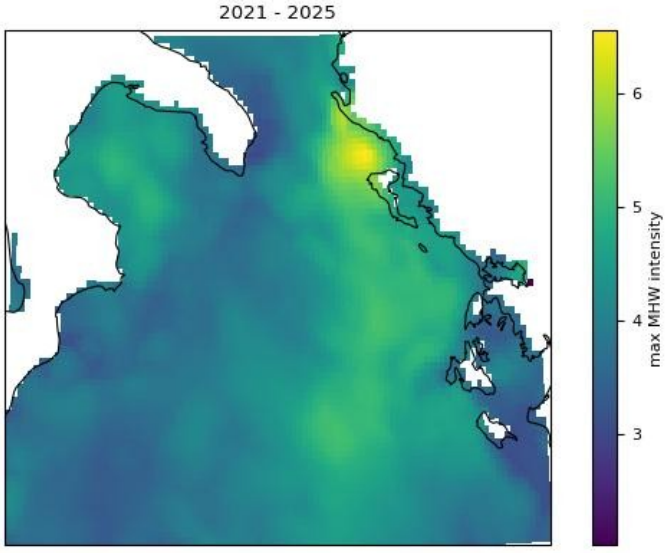


Map of maximum intensity

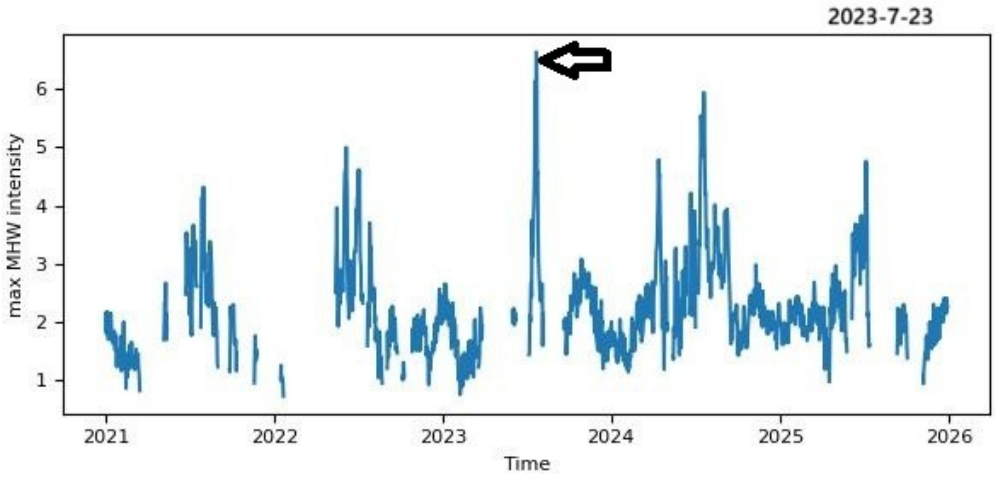
For the same subset (for the **2021–2025** period), in the **Figures** tab, we select the **max** statistic and click **PLOT**.



We obtain the map.

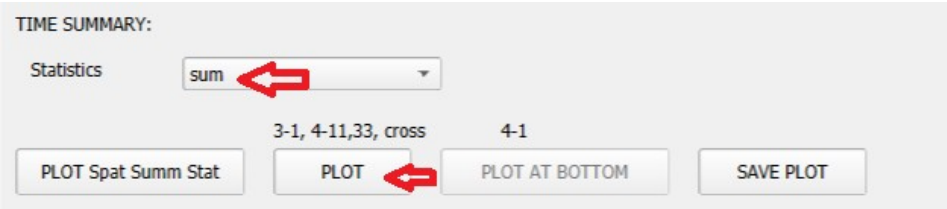


Additionally, using the **PLOT Spat Sum Stat** button allows us to determine when the maximum value occurred (using the cursor on the plot).

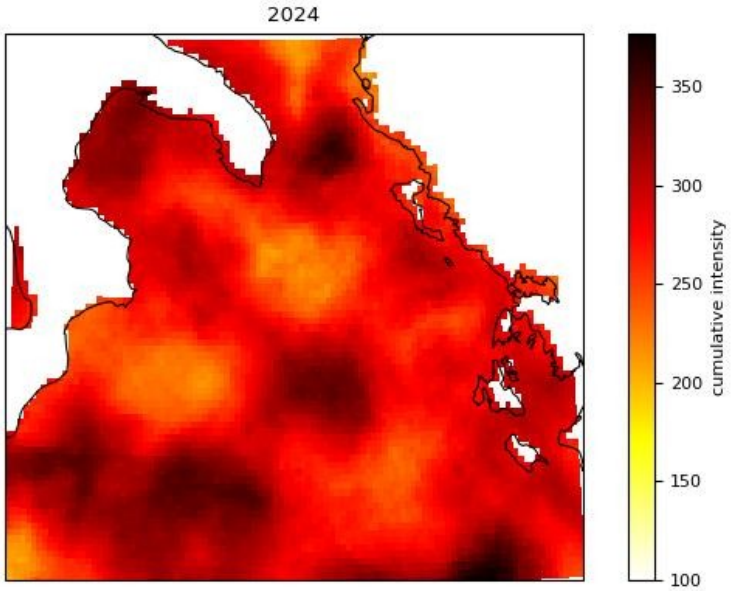


Map of cumulative intensity

The map will be created for **2024**, using the **anomaly** variable. In the **Statistics** tab, we set **Statistics** to *sum*. We sum the products of the number of days and intensity.

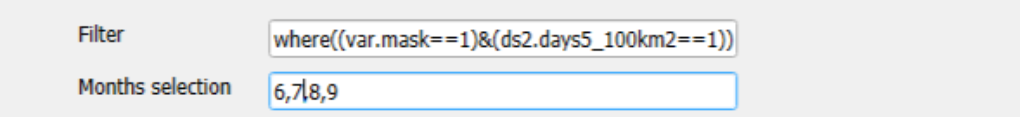


The resulting map is shown below.



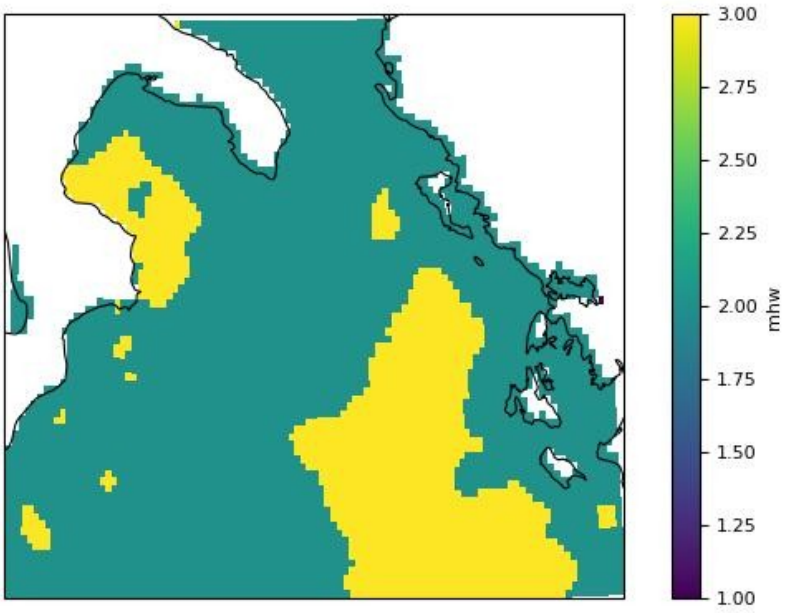
Map of maximum MHW category

The map will be created for a subset covering the entire period, but only for the summer months (June–September). When creating the subset for the **mhw** variable, we additionally specify the months included in the analysis.



We create the map in the **Statistics** tab by setting **Statistics** to *max*.

The resulting map will have the following form.



Object-based studies of MHW

Treat each MHW as a separate object defined in space and time.

We will use the file *".....\MHW_cat1_sumF.csv"*

created by filtering by minimum area (km²) and duration (days). The file describes each identified object and is sorted by area. The file contains **681 objects** and has the following form:

id,max_number_of_cells,date_start,date_end,date_of_max,days,area_km2

38729,6268,2023-07-11,2023-08-07,2023-07-24,28,151424.62142944336

27307,6268,2018-04-14,2018-06-21,2018-05-30,69,151424.62142944336

39981,6268,2023-12-21,2024-05-03,2024-04-08,135,151424.62142944336

42244,6268,2024-07-08,2025-04-18,2024-08-18,285,151424.62142944336

7118,6268,2002-03-18,2002-03-26,2002-03-22,9,151424.62142944336

5124,6268,2001-01-25,2001-04-13,2001-03-25,79,151424.62142944336

35853,6263,2022-06-19,2022-07-12,2022-07-03,24,151303.82961273193

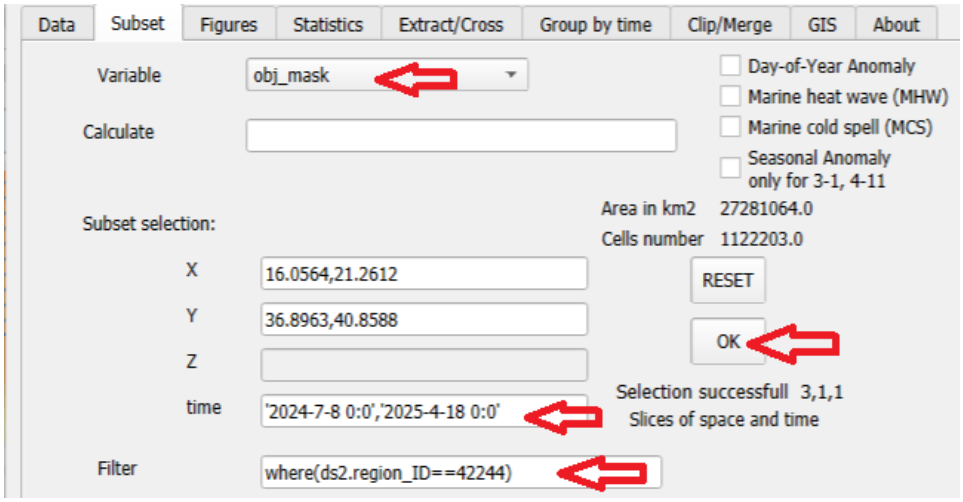
18675,6261,2013-04-24,2013-05-14,2013-05-03,21,151255.51288604736

.....

We will analyse two objects: the longest-lasting object and a significant object during the summer period.

We will start with a map showing the number of days for which object ID=42244 existed.

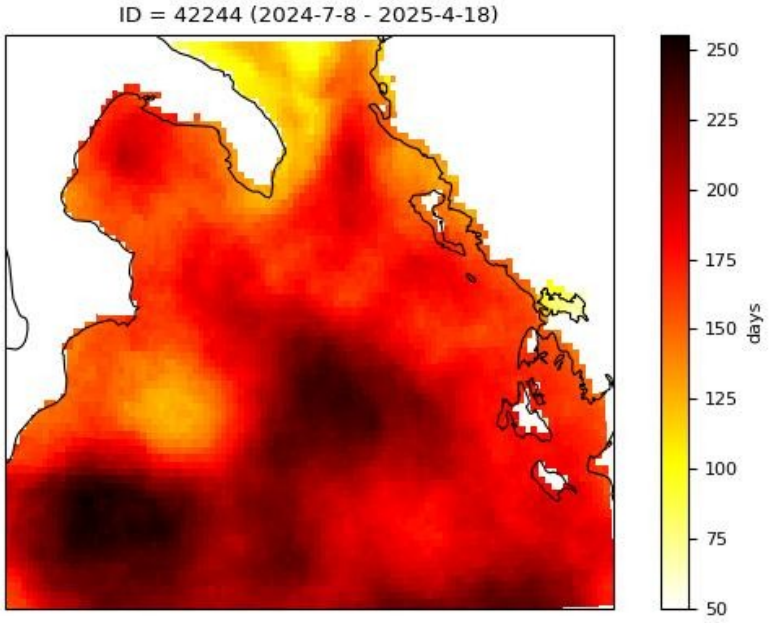
We use the variable, enter the time range (from the text file), and limit the subset to the selected object ID.



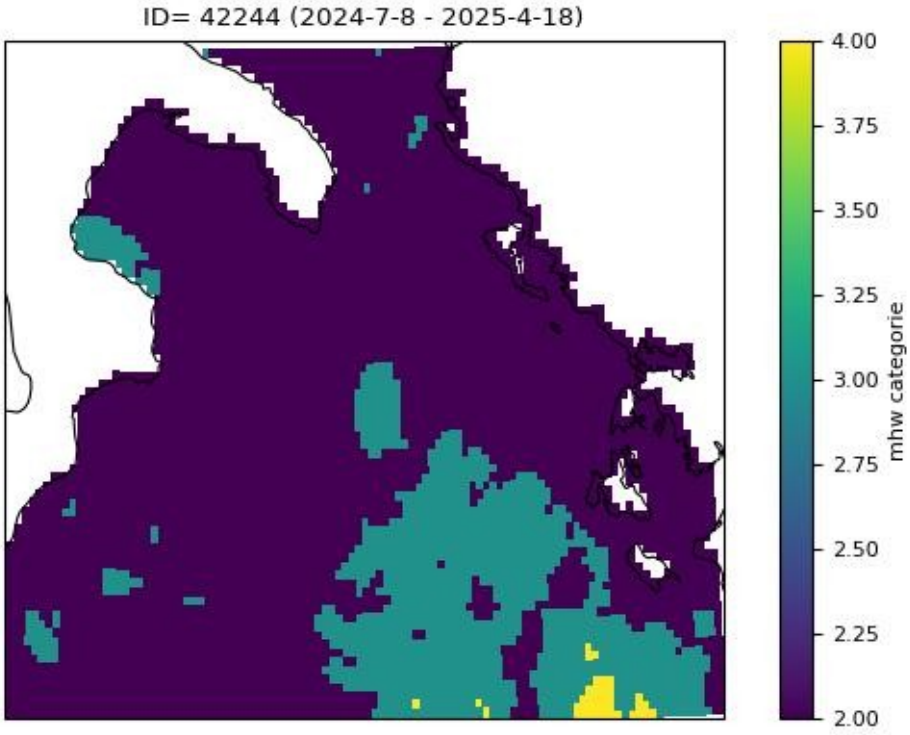
We switch to the **Statistics** tab.



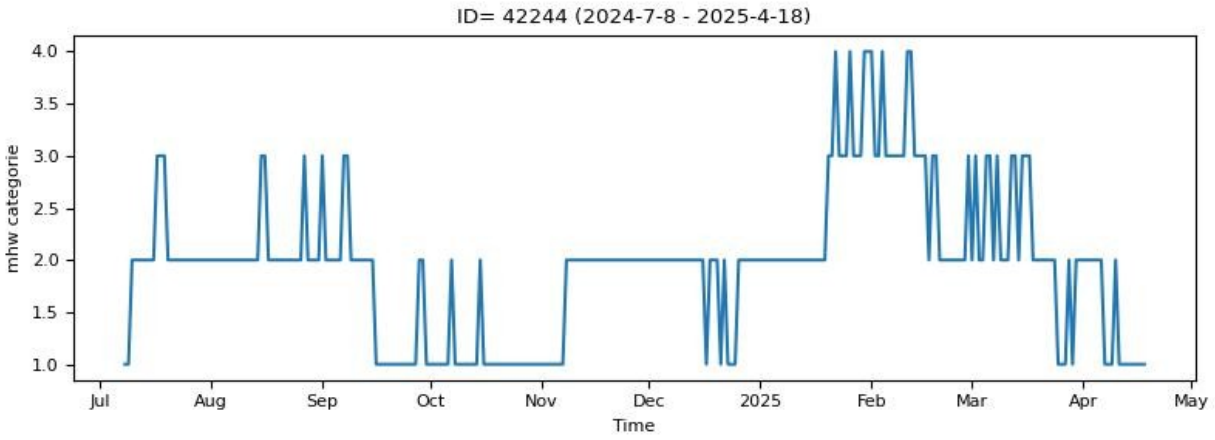
And we obtain the map.



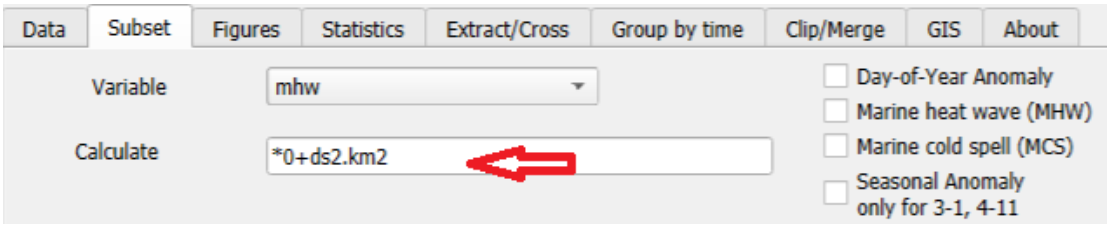
By changing the variable to **mhw** and the statistic to **max**, we can create a map of the maximum MHW categories.



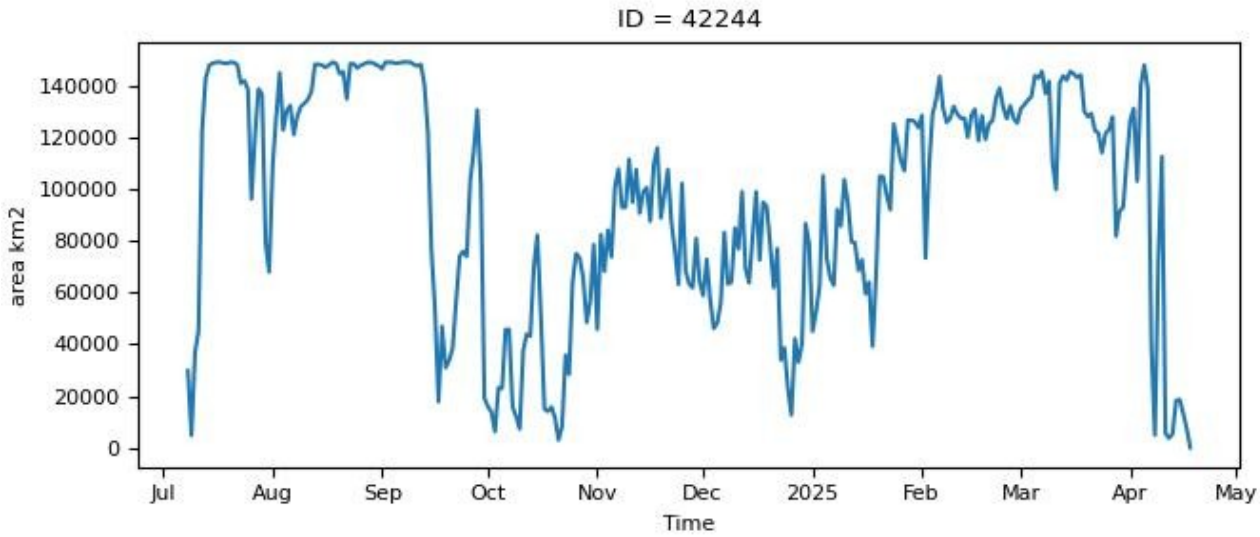
Then, by clicking **PLOT Spat Summ Stat**, we obtain a plot of the maximum MHW categories on successive days.



To create a plot showing changes in the MHW area for object ID=51900, we enter the variable `km2` in the **Calculate** field in the following form: (first multiplying any variable by zero).



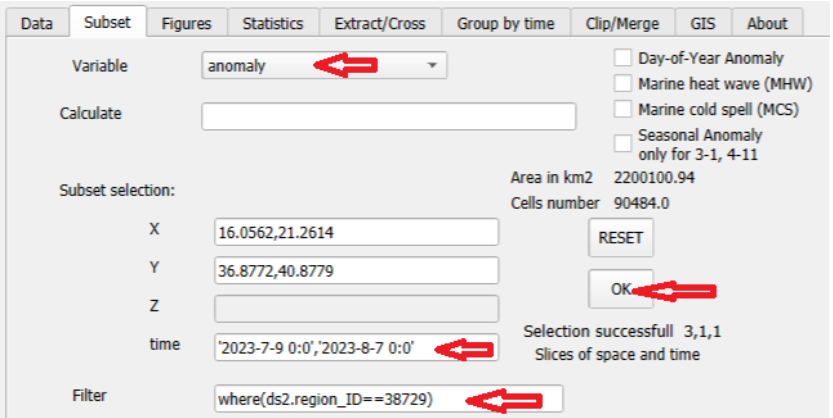
In the **Statistics** tab, we set *statistics* to **sum** and click **PLOT SpatSumm Stat**.



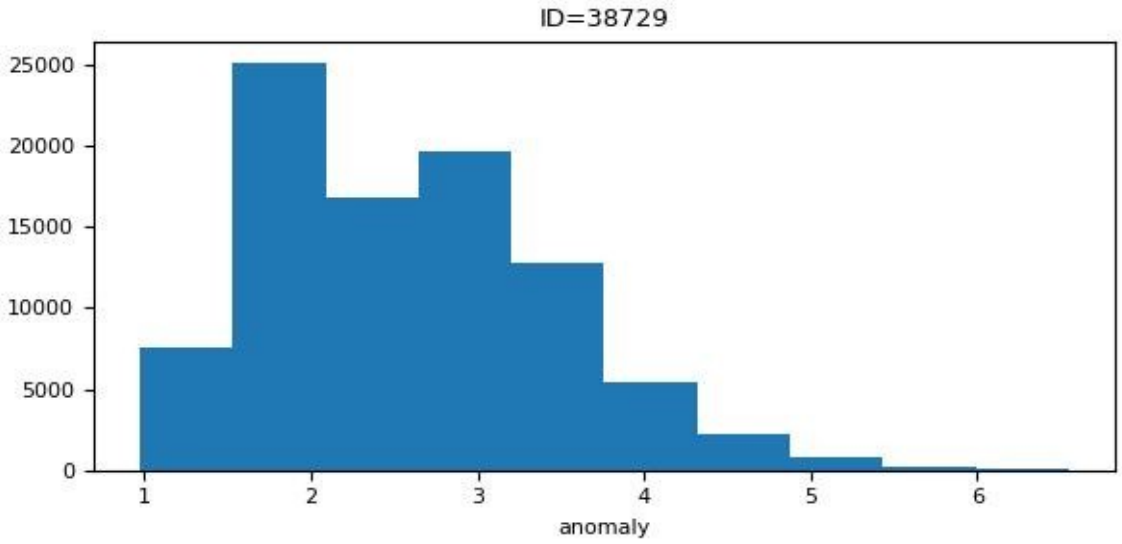
The basic parameters **mean intensity, maximum intensity, and cumulative intensity** for a specific object are determined as follows. We will analyze the following object:

38729,6268,2023-07-11,2023-08-07,2023-07-24,28,151424.62142944336

We create a subset for the `anomaly` variable and the selected object (ID and time range).



In the **Figures** tab, by clicking **PLOT**, we can create a histogram of the anomaly values for the analyzed MHW object.



In the **Statistics** tab, we have the following options:

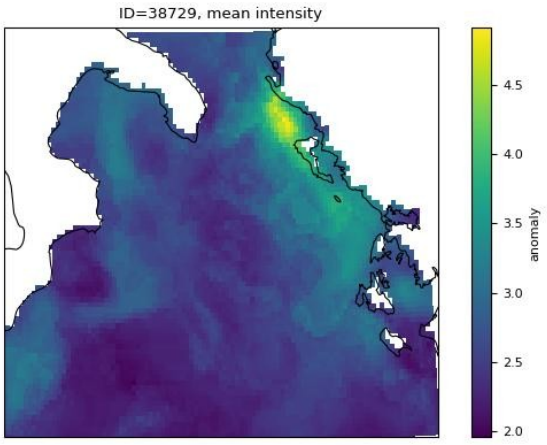
1. **Map of mean, maximum, and cumulative MHW intensity for ID=38729** — spatial distribution, with statistics calculated over time for each grid cell.

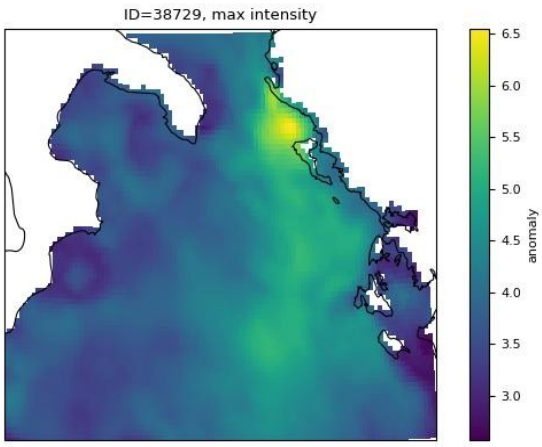
2. **Time series of mean and maximum MHW intensity for ID=38729** — temporal changes, with statistics calculated over the area of the group.
3. **A text file generated by the GIS Summary Statistics function**, providing a complete set of statistics for each day, calculated over the area of the group.

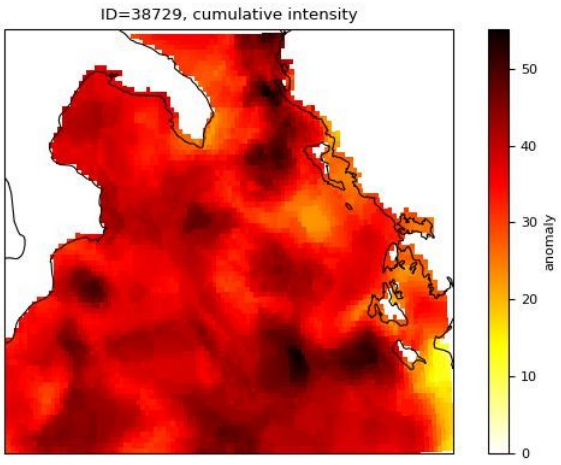
1. Map of mean, maximum, and cumulative MHW intensity for ID=38729 — spatial distribution, with statistics calculated over time for each grid cell.

In the **Statistics** tab, under **Time Summary**, we select the **mean, max, and sum** statistics and click **PLOT**.



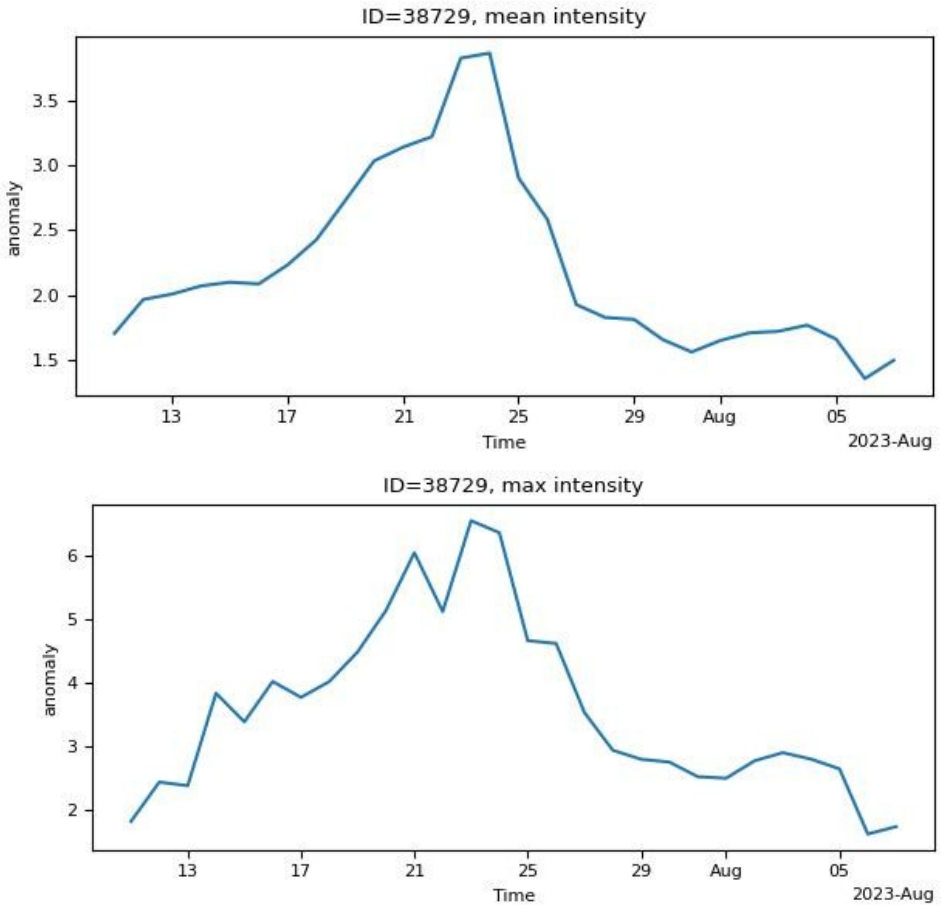






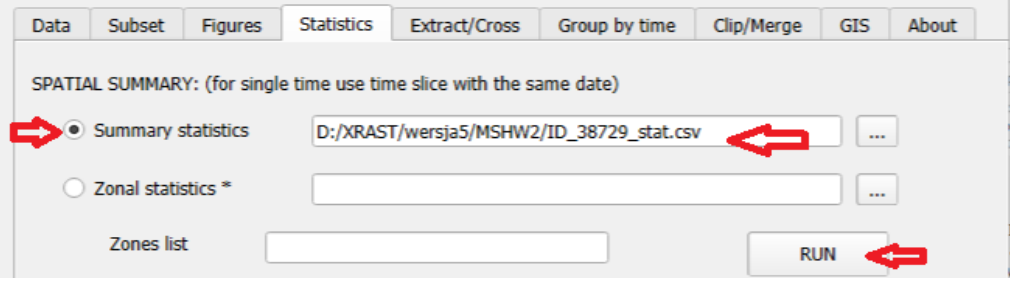
2. Time series of mean and maximum MHW intensity for ID=38729 — temporal changes, with statistics calculated over the area of the group.

In the **Statistics** tab, under **Time Summary**, we select the **mean** and **max** statistics and click **PLOT Spat Summ Stat**.



From these figures, we can see that the **maximum intensity** occurred on 23 July. This is an important parameter describing the group.

3. A text file generated by the GIS Summary Statistics function, providing a complete set of statistics for each day, calculated over the area of the group.



The resulting file has the following format:

time,mean,min,max,median,std,sum

2023-07-09 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-10 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-11 00:00:00,1.71,1.5726,1.8098,1.715,0.0536, 218.88

2023-07-12 00:00:00,2.0153,1.6343,2.4291,1.9888,0.2232, 326.49

2023-07-13 00:00:00,2.033,1.5925,2.3725,2.03,0.1928, 367.97

2023-07-14 00:00:00,2.1457,1.3309,3.8297,2.1154,0.4108, 6391.95

2023-07-15 00:00:00,2.0645,1.3333,3.3784,2.0449,0.4044, 7351.58

2023-07-16 00:00:00,2.0229,1.312,4.0145,2.0073,0.3212, 8302.05

2023-07-17 00:00:00,2.1752,1.3353,3.7651,2.0702,0.4534, 10793.56

.....

2023-08-01 00:00:00,1.6073,1.1109,2.4899,1.5971,0.1973, 4627.35

2023-08-02 00:00:00,1.6175,1.1201,2.7639,1.5665,0.2361, 4006.5

2023-08-03 00:00:00,1.6386,1.091,2.893,1.6061,0.2444, 4242.36

2023-08-04 00:00:00,1.6762,1.0586,2.7908,1.6471,0.3162, 5042.07

2023-08-05 00:00:00,1.5976,0.9724,2.6375,1.5515,0.2888, 3398.1

2023-08-06 00:00:00,1.3601,1.1288,1.6103,1.3662,0.085, 107.45

2023-08-07 00:00:00,1.5465,1.2704,1.7274,1.5773,0.121, 125.27

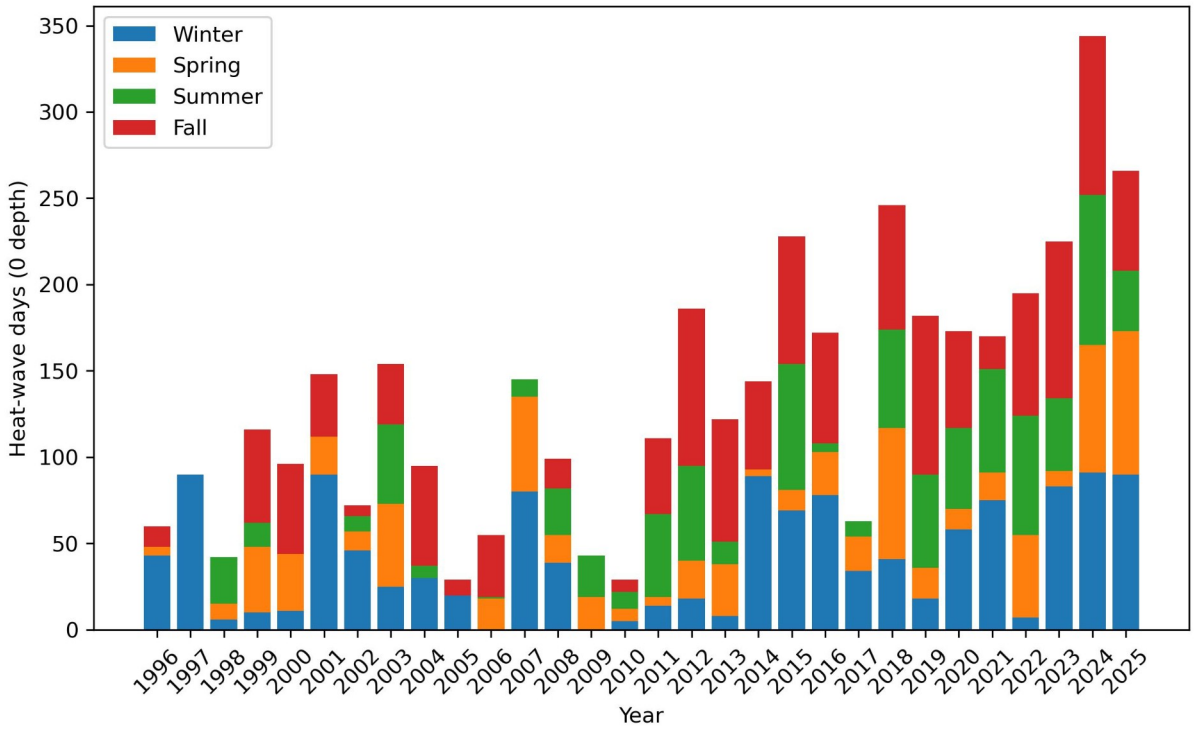
.....

MHW_extension2.ipynb is a Jupyter Notebook file (for Jupyter Notebook / Jupyter Lab).

This notebook contains scripts for creating additional MHW/MCS analysis products.

Script 1 (a, b)

Creates a stacked bar chart of MHW/MCS events. It shows the number of MHW/MCS days by season for each year.



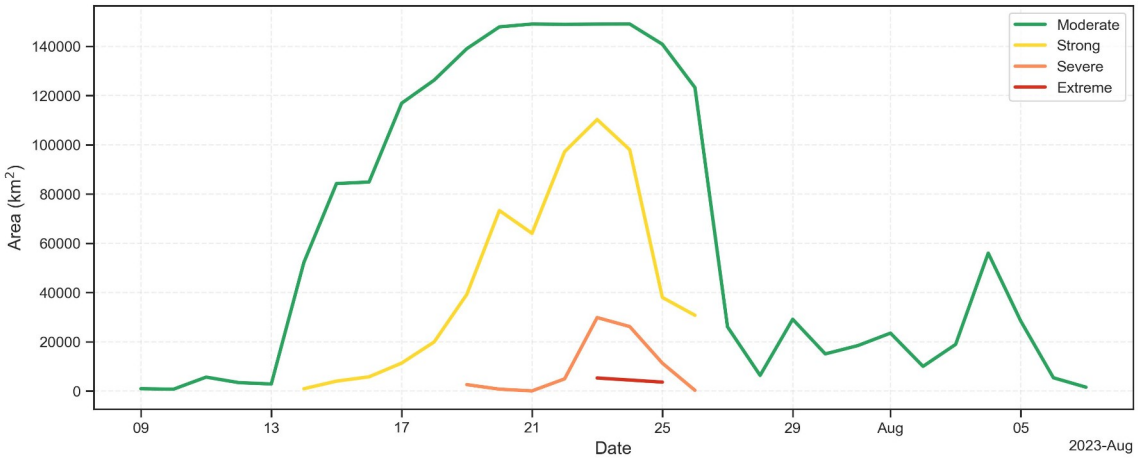
The figure is created in two steps.

In the first step (Script 1a), the file XXX is generated from MHW_cat1_sumF.csv. It contains a table of the total number of MHW/MCS days by season for each year.

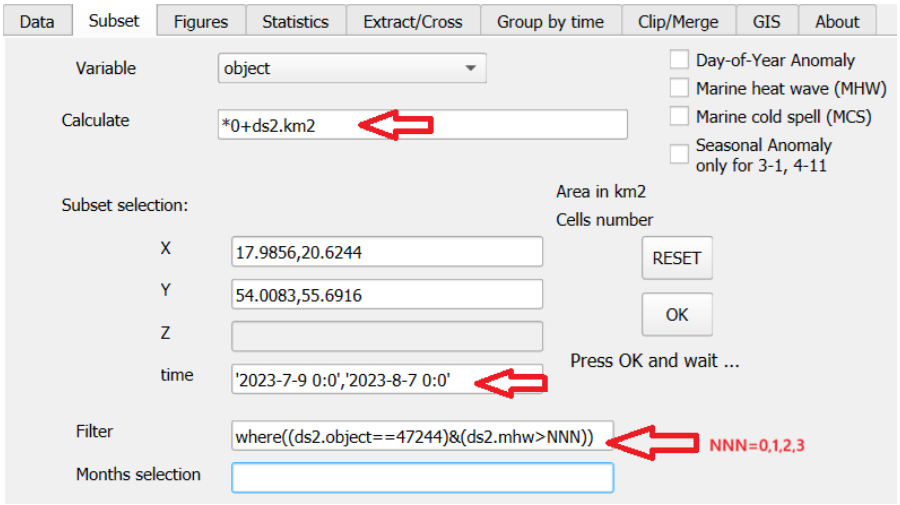
In the second step (Script 1b), the figure is generated from file XXX.

Script 2

Spatial extent (surface area) of MHWs or MCSs object in different categories, based on summary statistics.



The figure is created based on files (1–4) generated by the Summary Statistics function from the following SUBSETs:



where:

- Variable – any variable.
- Calculate – use the `km2` variable.
- Time – object duration (e.g. ID = 47244), taken from the MHW_cat1_sumF.csv file.
- Filter – depending on the number of categories present within the object area, the condition is set to NNN = 0, 1, 2, or 3.

The Summary Statistics function is then calculated separately for each category. The resulting files (2–4) are used as input parameters for this script.

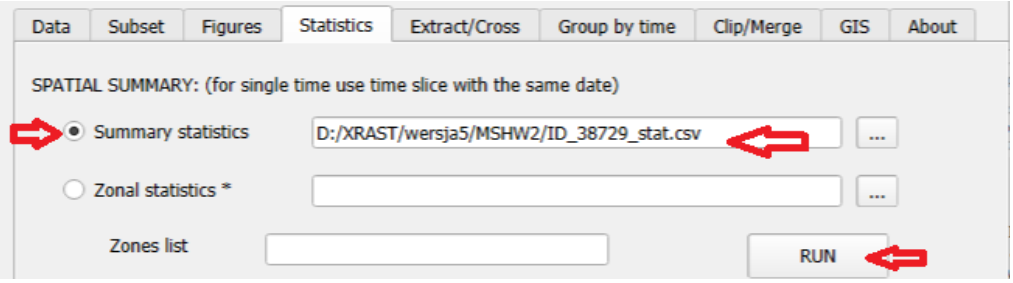
Script 3

Calculates basic statistics describing MHW/MCS events over the entire period, using the temperature anomaly variable from the file generated by the Summary Statistics function.

minimum intensity

maximum intensity

cumulative intensity



time,mean,min,max,median,std,sum

2023-07-09 00:00:00,nan,nan,nan,nan,nan, 0.0

2023-07-10 00:00:00,nan,nan,nan,nan,nan, 0.0

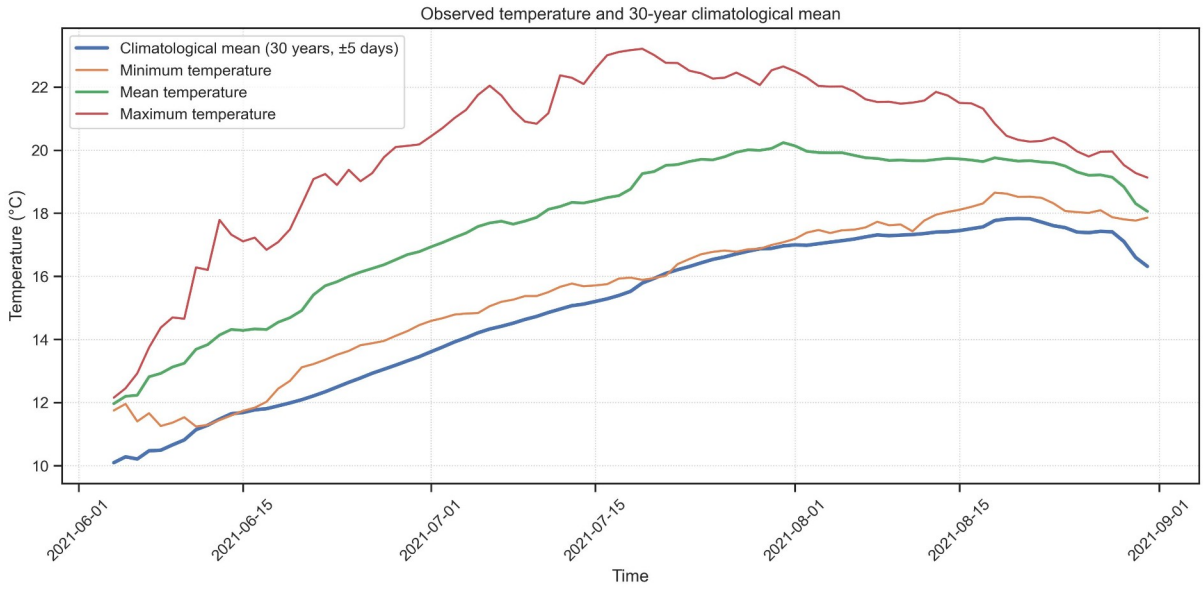
2023-07-11 00:00:00,1.71,1.5726,1.8098,1.715,0.0536, 218.88

2023-07-12 00:00:00,2.0153,1.6343,2.4291,1.9888,0.2232, 326.49

Script 4

Creates a figure showing the time series of four variables for a given object: climatological mean, minimum temperature, mean temperature, and maximum temperature.

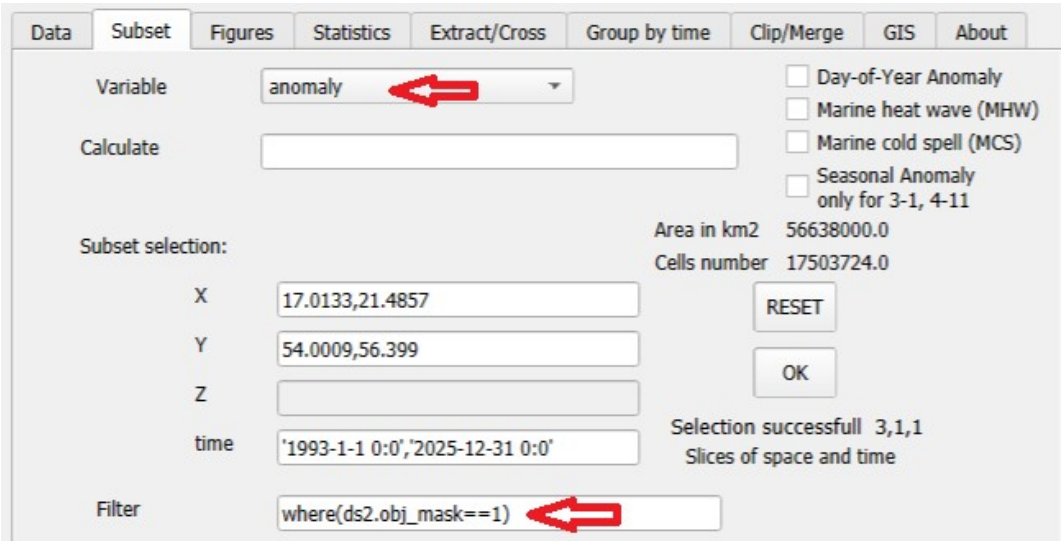
This is a typical figure illustrating how the temperature of a given object differs from the climatological mean.



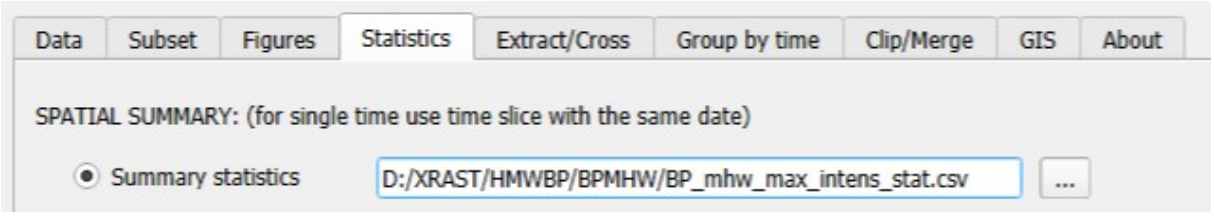
Script 5

Creates a figure showing the annual mean maximum intensity of MHWs/MCSs.

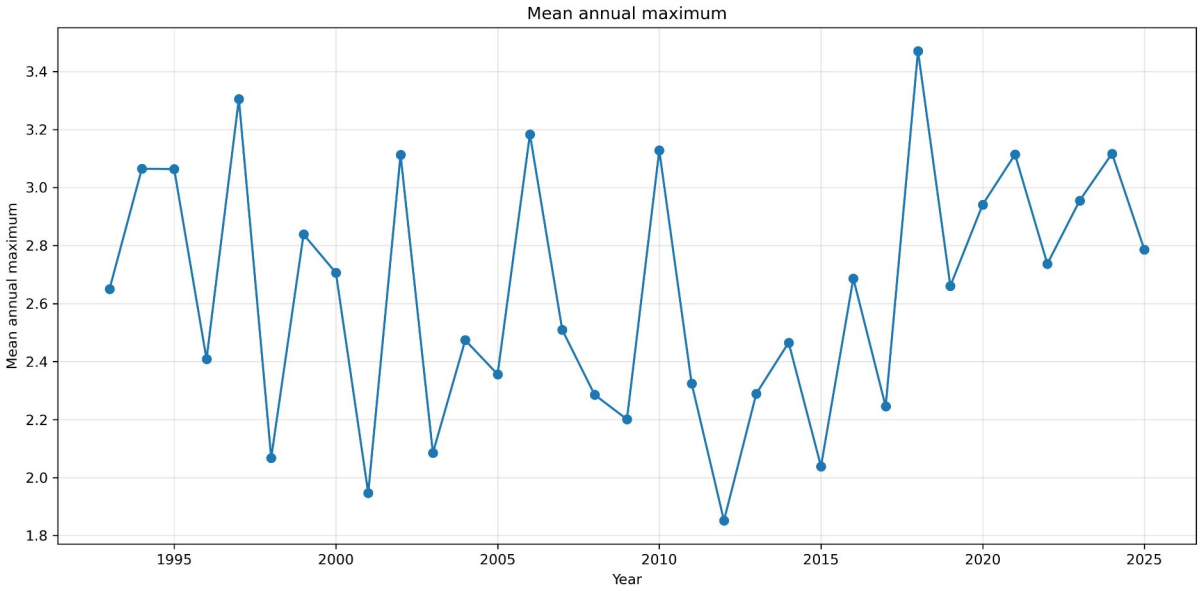
It is generated from a SUBSET in the form of...



For the anomaly variable, using a condition that selects only MHW and MCS areas. Summary Statistics is then calculated for the resulting SUBSET.



This file is used as an input parameter for the script, which generates the figure.



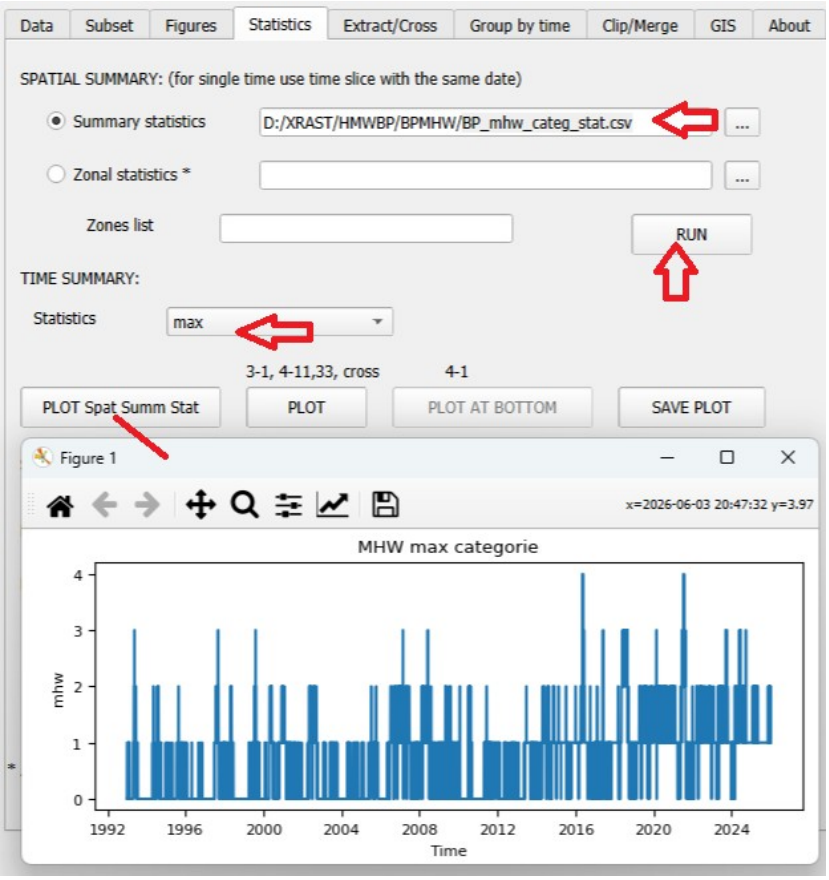
Script 6

Adds a column containing the category of each object to the text file ..._sumF.csv.

Contains three parameters:

1. The file ...._sumF.csv

2. The Summary Statistics file for the MHW/MCS variable (e.g. BP_mhw_categ_stat.csv), obtained from:



(The chart showing the maximum statistic has been added for illustrative purposes.

3. Output file, e.g. ...._sumF2.csv

The generated file has the following structure:

id,max_number_of_cells,date_start,date_end,date_of_max,days,area_km2,max_cat

79646,13456,2019-10-16,2020-06-24,2020-02-03,253,43951.87448120117,3

86326,13456,2020-09-21,2021-03-21,2020-11-19,182,43951.87448120117,2

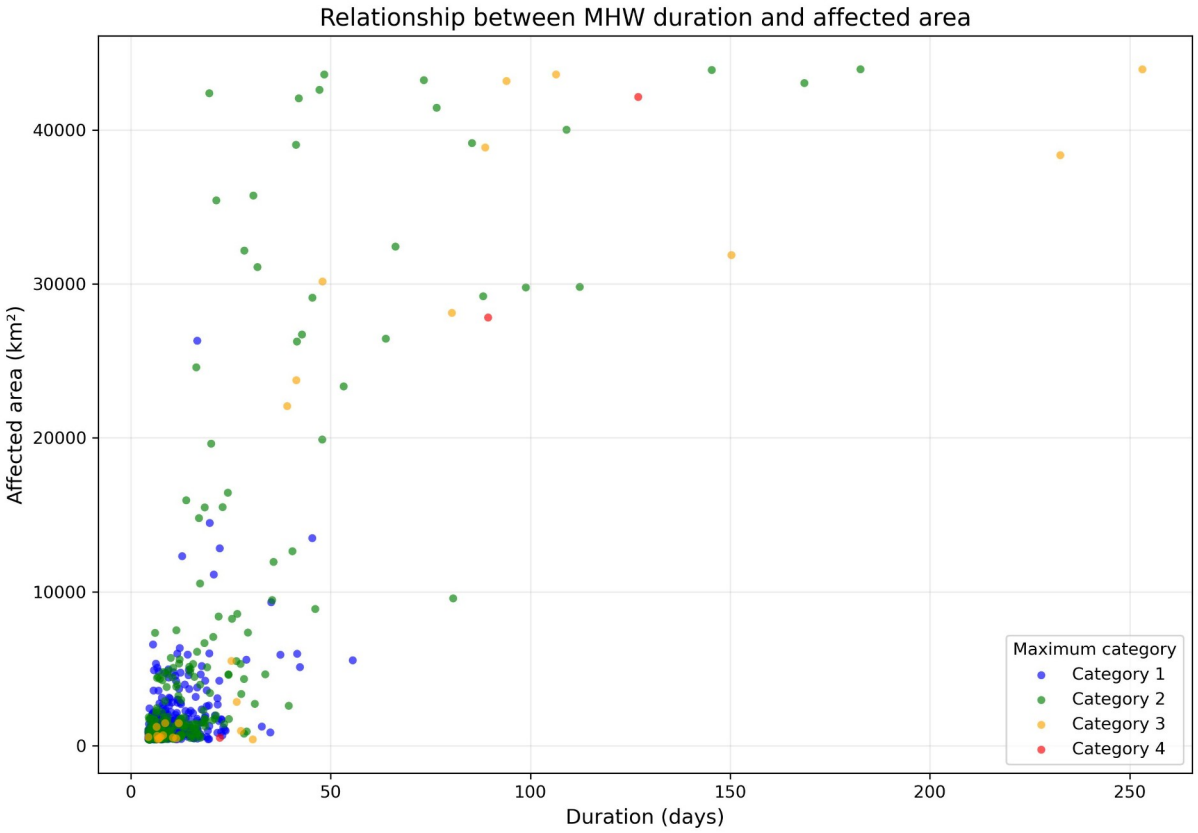
114762,13442,2024-12-22,2025-05-15,2025-04-04,145,43906.14571762085,2

22997,13355,2006-09-18,2006-11-04,2006-10-26,48,43621.9741153717,2

Script 7

Creates a figure showing the relationship between MHW object duration and their area.

The input parameter is the file generated by Script 6 (the file containing object descriptions, supplemented with the object category).



Reference

Hobday, A. J., Alexander, L. V., Perkins, S. E., Smale, D. A., Straub, S. C., Oliver, E. C. J., Benthuysen, J. A., Burrows, M. T., Donat, M. G., Feng, M., Holbrook, N. J., Moore, P. J., Scannell, H. A., Sen Gupta, A., & Wernberg, T. (2016). A hierarchical approach to defining marine heatwaves. *Progress in Oceanography, 141*, 227–238. **https://doi.org/10.1016/j.pocean.2015.12.014**. NOAA Institutional Repository

Hobday, A.J., L.V. Alexander, S.E. Perkins, D.A. Smale, S.C. Straub, E.C.J. Oliver, J.A. Benthuysen, M.T. Burrows, M.G. Donat, M. Feng, N.J. Holbrook, P.J. Moore, H.A. Scannell, A. Sen Gupta, and T. Wernberg. 2016. A hierarchical approach to defining marine heatwaves. *Progress in Oceanography* 141:227–238, https://doi.org/10.1016/j.pocean.2015.12.014. sciencedirect.com

Oliver, E.C.J., M.G. Donat, M.T. Burrows, P.J. Moore, D.A. Smale, L.V. Alexander, J.A. Benthuysen, M. Feng, A. Sen Gupta, A.J. Hobday, N.J. Holbrook, S.E. Perkins-Kirkpatrick, H.A. Scannell, S.C. Straub, and T. Wernberg. 2018. Longer and more frequent marine heatwaves over the past century. Nature Communications 9:1324, https://doi.org/10.1038/s41467-018-03732-9